



Interface reduction technique for Enhanced Craig-Bampton method

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ABSTRACT

Substructure coupling and model order reduction using Component Mode Synthesis (CMS) have, over recent years, gained considerable attention in the vibroacoustic analysis of complex structures. In the CMS methodology, the interior dynamics of each subcomponent in a substructured system are represented by a truncated set of normal modes within the lower frequency range, while all physical degrees of freedom (DOFs) at the interface are retained. In cases where there are many interconnected subcomponents within a system, in particular when these components are finely discretised in the Finite Element (FE) domain, the reduced system matrices may still involve a significant number of equations. This, in turn, leads to a considerable computational workload. To address this issue, further reduction of the system matrices concerning the interface DOFs by using a set of truncated interface modes can be considered. However, the accuracy of the reduced matrices depends on the representation of the truncated dynamics in the reduction process. In this work, two interface reduction techniques are presented to truncate the interface dynamics of the Enhanced Craig-Bampton (ECB) equations of motion. The first technique is a classical interface reduction approach that assumes decoupled internal and interface dynamics. The second approach is an extension of the first one by incorporating an additional coupling term that accounts for interactions between the truncated internal and interface dynamics. The performance of each interface reduction technique is evaluated by applying them to three practical engineering examples. In these instances, resonance frequencies, associated errors, transfer functions, and normal modes are compared to those obtained using both the classical CB method and the full model.

1. Introduction

Substructure coupling for dynamic analysis or Component Mode Synthesis (CMS) has, since the introduction of the Craig-Bampton (CB) method [1], gained great popularity within the field of Structural Dynamics. The CMS method is an extension of static condensation by Hurty [2,3] and Guyan-Irons [4,5], which in difference compensates for the neglected inertia terms by inclusion of a set of generalised coordinates. The objective of CMS comprises the fragmentation of a complex structure into several distinct regions or substructures, reducing the representation system matrices with contribution from component normal modes and assembling a reduced-order system with interface constraint (or attachment) modes for the entire system, Fig. 1. The constraint modes are defined as static deformation due to the application of a unit displacement at one interface degree-of-freedom (DOF), whilst restraining the remaining DOFs. Attachment modes are defined as displacement vectors due to the application of a unit force at one DOF, and hence are simply just columns of the associated flexibility matrix, [6]. Component normal modes are eigenvectors

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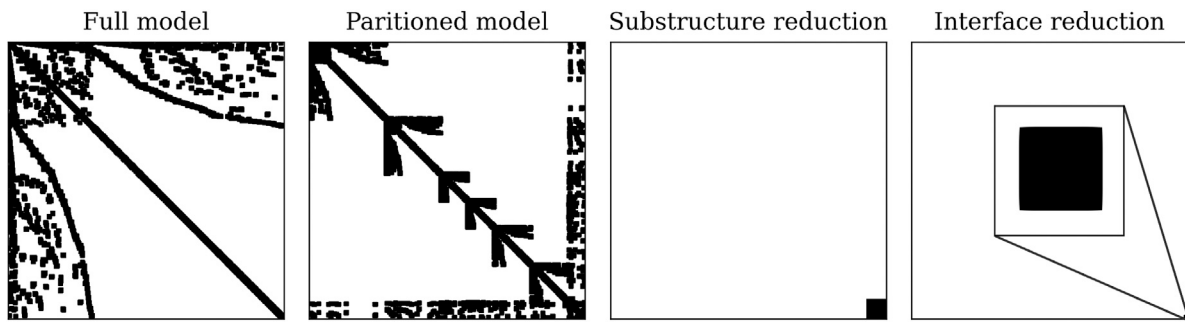


Fig. 1. Illustration of reduction process from the full model to interface reduction in the CMS domain. The black dots denote numbers associated with DOFs in the system matrix. The presented data is associated with the stiffened steel plate example in Section 4.1.

computed through eigenvalue analysis and may be classified as fixed-interface, free-interface, hybrid-interface and loaded interface normal modes, [7]. The number of component normal modes (dominant modes) are often selected within the lower frequency range or chosen based on mode selection methods, for example, [8–10]. Further review of substructure coupling and CMS methods can be found in [1,7,11,12].

The Craig-Bampton method employs a truncated set of fixed-interface normal modes together with a set of constraint modes to reduce the substructure system matrices. The fixed-interface modes are computed through an eigenvalue analysis for the substructure system matrices with restrained interface boundary DOFs. In the 1970s, another class of the CMS method was introduced to employ free-interface modes, in particular MacNeal's Method [13], Rubin's Method [14], Hintz's method [15] and Craig-Chang Method [16], and others [17–20]. Substructures with truncated internal dynamics are assembled into a coupled global system using displacement compatibility at their interfaces, referred to as primal assembly, [11,21]. Considering force compatibility at the substructure interfaces, Rixen [21] introduced a Dual Assembly approach allowing for small discontinuities in displacement across the interfaces. Park and Park also proposed a robust free interface CMS method known as flexibility-based CMS (F-CMS) with localised Lagrange multipliers to ensure mathematical independence between substructures [22]. The present study focuses on the primal assembly approach first introduced by Craig Jr. and Bampton [1]. This approach considers the coupling of the substructures at their interfaces through the use of constraint modes, thereby allowing for the description of the motion at the coupling interfaces through these constraint modes.

The adoption of reduced substructural models instead of full models in computational mechanics results in improved computational efficiency, simplified modelling, and increased flexibility. Hence, in addition to its use in structural mechanics to study mechanical response of structures, the CMS method has been recently extended to address computational efficiency in fluid–structure interaction, thermo-mechanical vibration, vibro-acoustic analyses and simulation of protein dynamics, as reported in [23–26]. Accordingly, reduction methods, particularly the CB method, have been integrated into commercial FE software, [27,28].

The accuracy of CMS methods, including the CB method, is often contingent upon the number of component normal modes included in the compensation for neglected inertia. To address this limitation, Kim and Lee [19] developed the Enhanced Craig-Bampton (ECB) method, which incorporates both dominant and residual modes truncated in the original CB method. This results in a significant improvement in the accuracy of the CB method with a minimal additional computational cost. Accurate reduced models are crucial for structural design, system identification, and model calibration based on experimental data.

The reduction process within the CMS framework involves truncation of the internal dynamics of each subcomponent within a substructured system using a set of dominant modes, whilst preserving the physical DOFs at the interfaces between the substructures. This procedure is particularly useful in the analysis of structural assemblies consisting of a limited number of interconnected substructures or with a limited number of interaction nodes between the substructures. Nonetheless, for extremely large and complex structures with multiple substructures and a substantial number of interaction nodes, the CMS domain may be dominated by the interface DOFs. As a result, the reduced-order system matrices may still contain a significant number of DOFs, leading to computational costs. In 1977, Craig and Chang introduced the interface reduction technique to further reduce the number of DOFs by applying Guyan, Ritz, and modal reduction techniques, as documented in [29]. The modal reduction technique underwent further development by Castanier et al. [30], who applied a secondary eigenvalue analysis on the system-level interface partition of the assembled system matrices. This led to the identification of eigenvectors known as system-level Characteristic Constraint (CC) modes. Using a truncated set of CC modes allows for further reduction of the equations of motion and results in a significant decrease in the size of the assembled system matrices.

The work by Castanier et al. [30] inspired subsequent studies in the field of interface reduction technique. For example, Rixen proposed an interface reduction technique for the Dual Craig-Bampton method utilising free interface modes [21]. Holzwarth et al. [31] proposed the utilisation of Legendre Polynomials as basis functions to represent the interface deformation patterns as an approach to reducing the interface DOFs. The performance of this method was then compared to that of the CC mode approach. Ahn et al. [32] presented a refinement technique of the interface reduction technique by Castanier et al. [30] by incorporating the influence of residual flexibility into the CC modes. Additionally, interface reduction of the F-CMS method was proposed by using mathematical basis reconstruction of the interface DOFs, [17,33]. Hong et al. [34] proposed a technique for conducting eigenvalue

analysis on the CB interface DOFs at the substructure level before assembly using exact interface compatibility, and reducing the interface DOFs by employing augmented modes. Given the popularity of the CB method, most of the existing interface reduction techniques are associated with it. Further information on these techniques can be found in a review by Krattiger et al. [35].

The interface reduction technique using CC modes results in reduced-order system matrices, the accuracy of which depends on the number of interface constraint modes retained. However, the reduction process may result in system matrices of lower accuracy compared to the original system matrices due to approximations introduced during the reduction procedure. To mitigate this issue, the interface reduction technique based on CC modes can be combined with the ECB method, instead of CB. The ECB method has been shown to produce system matrices of greater accuracy compared to the original CB method, [19,36–38]. The approach to the interface reduction proposed by Castanier et al. [30] encompass a classical modal truncation based on the assumption of decoupled internal and interface DOFs. This approach is valid, considering the decoupled stiffness characteristics of the CB system matrices. However, as evidenced further in the paper, this assumption is not entirely correct and the internal and interface DOFs are not fully decoupled when the ECB method is employed. This may result in system matrices with reduced accuracy. Incorporating the coupling between internal and interface dynamics can improve the accuracy of the interface reduction technique and produce system matrices with greater accuracy compared to the classical approach proposed by Castanier et al. [30].

In this study, two interface reduction techniques are coupled with the ECB method. The first option, as proposed by Castanier et al. [30], is considered. The second option builds upon the first approach and includes a coupling term to address the transmission of vibrational energy between the internal and interface dynamics. The efficacy of each of these interface reduction techniques has been evaluated through three practical engineering examples, and the results demonstrate an improvement in the accuracy of the system matrices. Furthermore, the results are compared with the classical CB method, both with and without the interface reduction proposed by Castanier et al. [30]. Further mathematical derivations and a step-by-step implementation algorithm are also provided to facilitate the practical application of this work.

The structure of the rest of this paper is outlined as follows: Section 2 provides a comprehensive overview of substructure coupling and reduction of internal dynamics of substructures. Section 3 outlines the derivation of interface reduction techniques in conjunction with CB and ECB. The performance of each proposed interface reduction technique is evaluated through 3 examples in Section 4, and conclusions are drawn in Section 5.

2. Substructure coupling and model order reduction

This section provides a brief introduction to substructure reduction and coupling, comprising substructure partitioning, internal dynamics truncation, and subsequent coupling. The reduction methods used to truncate substructure internal dynamics include the Craig-Bampton (CB) and Enhanced Craig-Bampton methods. In the CMS domain, substructures are assembled block-diagonally and share interfaces, Fig. 1. With the aim of achieving a partitioned global system, the substructures are decomposed into internal and interface regions before the coupling procedure.

The equations of motion for a discretised substructure, in the absence of damping, can be formulated as:

$$\mathbf{M}^{(s)}\ddot{\mathbf{u}}^{(s)} + \mathbf{K}^{(s)}\mathbf{u}^{(s)} = \mathbf{f}^{(s)}, \quad s = 1, 2, \dots, N_s, \quad (1)$$

where $\mathbf{M}^{(s)}, \mathbf{K}^{(s)} \in \mathbb{R}^{N^{(s)} \times N^{(s)}}$ are mass and stiffness matrices and $\mathbf{u}^{(s)}, \mathbf{f}^{(s)} \in \mathbb{R}^{N^{(s)} \times 1}$ are nodal displacement and force vectors for substructure s , respectively. An upper double-dot denotes second-order time-derivative d^2/dt^2 and N_s denotes the total number of substructures. To simplify subsequent substructure coupling, the substructure's internal and boundary regions are decomposed to

$$\mathbf{M}^{(s)} = \begin{bmatrix} \mathbf{M}_{ii}^{(s)} & \mathbf{M}_{ib}^{(s)} \\ \mathbf{M}_{bi}^{(s)} & \mathbf{M}_{bb}^{(s)} \end{bmatrix}, \quad \mathbf{K}^{(s)} = \begin{bmatrix} \mathbf{K}_{ii}^{(s)} & \mathbf{K}_{ib}^{(s)} \\ \mathbf{K}_{bi}^{(s)} & \mathbf{K}_{bb}^{(s)} \end{bmatrix}, \quad \mathbf{u}^{(s)} = \begin{Bmatrix} \mathbf{u}_i^{(s)} \\ \mathbf{u}_b^{(s)} \end{Bmatrix}, \quad \mathbf{f}^{(s)} = \begin{Bmatrix} \mathbf{f}_i^{(s)} \\ \mathbf{f}_b^{(s)} \end{Bmatrix}, \quad (2)$$

where the substructure system matrices $\mathbf{M}^{(s)}$ and $\mathbf{K}^{(s)}$ are symmetric positive semi-definite. Hence, the system matrix blocks $\mathbf{M}_{ib}^{(s)} = \mathbf{M}_{bi}^{(s)T}$ and $\mathbf{K}_{ib}^{(s)} = \mathbf{K}_{bi}^{(s)T}$. The subscripts i and b denote internal and boundary (interface) regions, respectively. The total number of substructure internal and boundary DOFs are defined as $N_i^{(s)}$ and $N_b^{(s)}$, respectively. Thus, the total number of substructure DOFs is $N^{(s)} = N_i^{(s)} + N_b^{(s)}$.

The system matrices and force vector associated with N_s substructures to be coupled can be assembled in a block-diagonal format as follows:

$$\tilde{\mathbf{M}} \stackrel{\Delta}{=} \text{diag}(\mathbf{M}^{(1)}, \dots, \mathbf{M}^{(N_s)}) = \begin{bmatrix} \mathbf{M}^{(1)} & & \\ & \ddots & \\ & & \mathbf{M}^{(N_s)} \end{bmatrix}, \quad \tilde{\mathbf{K}} \stackrel{\Delta}{=} \text{diag}(\mathbf{K}^{(1)}, \dots, \mathbf{K}^{(N_s)}), \quad \tilde{\mathbf{f}} \stackrel{\Delta}{=} \begin{Bmatrix} \mathbf{f}^{(1)} \\ \vdots \\ \mathbf{f}^{(N_s)} \end{Bmatrix}. \quad (3)$$

In the CMS field, substructures in a coupled system share their boundary DOFs at their interfaces. To ensure compatibility between the coupled substructure DOFs, the Boolean matrix $\mathbf{L}$ is used to impose interface coupling between the substructures, as described in [11]. This establishes a relationship between the substructure displacement vector $\mathbf{u}^{(s)}$ and the global displacement vector $\mathbf{u}_g$ as follows:

$$\mathbf{u}^{(s)} = \mathbf{L}^{(s)}\mathbf{u}_g, \quad \mathbf{u}_g = \begin{Bmatrix} \mathbf{u}_i \\ \mathbf{u}_b \end{Bmatrix}, \quad (4)$$

where $\mathbf{L}^{(s)}$ is a substructure primal assembly Boolean operator. This relationship ensures the recovery of the substructure displacement vector from the global displacement vector. Considering a primal assembly approach, the substructures in Eq. (3) can be coupled as follows:

$$\mathbf{M}_g \stackrel{\Delta}{=} \mathbf{L}^T \tilde{\mathbf{M}} \mathbf{L} = \begin{bmatrix} \mathbf{M}_{ii} & \mathbf{M}_{ib} \\ \mathbf{M}_{bi} & \mathbf{M}_{bb} \end{bmatrix}, \quad (5a)$$

$$\mathbf{K}_g \stackrel{\Delta}{=} \mathbf{L}^T \tilde{\mathbf{K}} \mathbf{L} = \begin{bmatrix} \mathbf{K}_{ii} & \mathbf{K}_{ib} \\ \mathbf{K}_{bi} & \mathbf{k}_{bb} \end{bmatrix}, \quad (5b)$$

$$\mathbf{F}_g \stackrel{\Delta}{=} \mathbf{L}^T \tilde{\mathbf{F}} = \begin{Bmatrix} \mathbf{f}_i \\ \mathbf{f}_b \end{Bmatrix}, \quad (5c)$$

where the size of the global mass matrix $\mathbf{M}_g$, stiffness matrix $\mathbf{K}_g$ and force vector $\mathbf{F}_g$ is $N \times N$, $N \times N$, and $N \times 1$, respectively. The total number of DOFs is denoted $N = N_i + N_b$, where N_i represents the total number of internal DOFs, and N_b represents the total number of interface DOFs in the global system. The total number of internal DOFs is defined as $N_i = \sum_{s=1}^{N_s} N_i^{(s)}$. Because the substructures share interfaces within the global system, the total number of global interface DOFs is less than the sum of all individual substructure interface DOFs, i.e., $N_b < \sum_{k=1}^{N_s} N_b^{(s)}$.

The dimensions of $\mathbf{L}$, $\mathbf{u}^{(s)}$, and $\mathbf{u}_g$ are $N_b \times N$, $N^{(s)} \times 1$, and $N \times 1$, respectively. The global force vector $\mathbf{F}_g$ contains a zero partition for $\mathbf{f}_i$ since all loaded DOFs are considered as interface DOFs and included in the $\mathbf{f}_b$ partition. Information on constructing the Boolean matrix $\mathbf{L}$ can be found in [11]. It is important to note that these equations represent the coupling equations, originally proposed by Craig Jr. and Bampton [1], and referred to as primal assembly by Rixen [39]. The next section will provide a brief review of the CB and ECB methods for reducing the substructure internal DOFs before the assembly procedure outlined in Eq. (5).

2.1. Reduction of internal DOFs

The internal DOFs of each substructure can be reduced by utilising a set of component fixed-interface normal modes before the assembly and coupling procedure outlined in Eqs. (3) and (5). The normal modes are obtained by restraining all interface DOFs and conducting an eigenvalue analysis for the internal DOFs as:

$$\left(\mathbf{K}_{ii}^{(s)} - \omega_{i,j}^{(s)2} \mathbf{M}_{ii}^{(s)} \right) \boldsymbol{\phi}_{i,j}^{(s)} = \mathbf{0}, \quad j = 1, 2, \dots, N_{i,d}^{(s)}, \quad (6)$$

where the system matrices $\mathbf{M}_{ii}^{(s)}$ and $\mathbf{K}_{ii}^{(s)}$ are respectively internal partitions of the substructure mass and stiffness matrices outlined in Eq. (2), whilst $\omega_{i,j}^{(s)2}$ and $\boldsymbol{\phi}_{i,j}^{(s)}$ are associated j th eigensolution. The number of substructure dominant modes retained is defined as $N_{i,d}^{(s)}$. The dominant modes are often selected within the lower frequency range, or according to mode selection techniques, [9,10].

Considering a fixed-interface truncation approach, the substructure displacement vector can be approximated to

$$\mathbf{u}^{(s)} \approx \tilde{\mathbf{T}}_{\text{CB}}^{(s)} \hat{\mathbf{u}}^{(s)} = \tilde{\mathbf{T}}_{\text{CB}}^{(s)} \begin{Bmatrix} \mathbf{q}_{i,d}^{(s)} \\ \mathbf{u}_b^{(s)} \end{Bmatrix}, \quad (7a)$$

$$\tilde{\mathbf{T}}_{\text{CB}}^{(s)} = \begin{bmatrix} \boldsymbol{\Phi}_{i,d}^{(s)} & \boldsymbol{\Psi}_{ib}^{(s)} \\ \mathbf{0} & \mathbf{I}_{bb}^{(s)} \end{bmatrix}, \quad \boldsymbol{\Phi}^{(s)} = \begin{bmatrix} \boldsymbol{\phi}_{i,1}^{(s)} & \boldsymbol{\phi}_{i,2}^{(s)} & \dots & \boldsymbol{\phi}_{i,d}^{(s)} \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} \end{bmatrix}, \quad \boldsymbol{\Psi}^{(s)} = \begin{bmatrix} \boldsymbol{\Psi}_{ib}^{(s)} \\ \mathbf{I}_{bb}^{(s)} \end{bmatrix}, \quad \boldsymbol{\Psi}_{ib}^{(s)} = -\mathbf{K}_{ii}^{(s)-1} \mathbf{K}_{ib}^{(s)}, \quad (7b)$$

where the substructure displacement vector $\mathbf{u}^{(s)}$ is approximated to $\hat{\mathbf{u}}^{(s)}$, by making use of the substructure transformation matrix $\tilde{\mathbf{T}}_{\text{CB}}^{(s)}$, where an over-bar denotes approximated quantities. The fixed-interface mode matrix $\boldsymbol{\Phi}^{(s)}$ contains $N_{i,d}$ columns of the dominant modes obtained in Eq. (6). The constraint mode matrix $\boldsymbol{\Psi}^{(s)}$ contains static mode shapes of the internal DOFs due to unit displacement of the interface boundary DOFs $\boldsymbol{\Psi}_{ib}^{(s)}$ and an identity matrix $\mathbf{I}_{bb}^{(s)}$. Thus, the motion of the interface is entirely described by the constraint modes, [30]. The $\boldsymbol{\Phi}^{(s)}$ together with $\boldsymbol{\Psi}^{(s)}$ are referred to as a complete set of fixed-interface normal mode and constraint mode and have the dimensions $N \times N_{i,d}$ and $N \times N_b$, respectively. Truncating the substructure internal DOFs using Eq. (7) in Eq. (1) and pre-multiplying by $\tilde{\mathbf{T}}_{\text{CB}}^{(s)T}$, the reduced substructure equations of motion can be expressed as:

$$\tilde{\mathbf{T}}_{\text{CB}}^{(s)T} \mathbf{M}^{(s)} \tilde{\mathbf{T}}_{\text{CB}}^{(s)} \begin{Bmatrix} \hat{\mathbf{q}}_{i,d}^{(s)} \\ \hat{\mathbf{u}}_b^{(s)} \end{Bmatrix} + \tilde{\mathbf{T}}_{\text{CB}}^{(s)T} \mathbf{K}^{(s)} \tilde{\mathbf{T}}_{\text{CB}}^{(s)} \begin{Bmatrix} \mathbf{q}_{i,d}^{(s)} \\ \mathbf{u}_b^{(s)} \end{Bmatrix} = \tilde{\mathbf{T}}_{\text{CB}}^{(s)T} \begin{Bmatrix} \mathbf{f}_i^{(s)} \\ \mathbf{f}_b^{(s)} \end{Bmatrix}, \quad (8)$$

with

$$\hat{\mathbf{M}}^{(s)} = \tilde{\mathbf{T}}_{\text{CB}}^{(s)T} \mathbf{M}^{(s)} \tilde{\mathbf{T}}_{\text{CB}}^{(s)} = \begin{bmatrix} \hat{\mathbf{M}}_{ii}^{(s)} & \hat{\mathbf{M}}_{ib}^{(s)} \\ \hat{\mathbf{M}}_{bi}^{(s)} & \hat{\mathbf{M}}_{bb}^{(s)} \end{bmatrix}, \quad (9a)$$

$$\hat{\mathbf{K}}^{(s)} = \tilde{\mathbf{T}}_{\text{CB}}^{(s)T} \mathbf{K}^{(s)} \tilde{\mathbf{T}}_{\text{CB}}^{(s)} = \begin{bmatrix} \hat{\mathbf{K}}_{ii}^{(s)} & \mathbf{0}_{ib}^{(s)} \\ \mathbf{0}_{bi}^{(s)} & \hat{\mathbf{K}}_{bb}^{(s)} \end{bmatrix}, \quad (9b)$$

$$\hat{\mathbf{F}}^{(s)} = \tilde{\mathbf{T}}_{\text{CB}}^{(s)T} \begin{Bmatrix} \mathbf{f}_i^{(s)} \\ \mathbf{f}_b^{(s)} \end{Bmatrix} = \begin{Bmatrix} \hat{\mathbf{f}}_i^{(s)} \\ \hat{\mathbf{f}}_b^{(s)} \end{Bmatrix}, \quad (9c)$$

where $\hat{\mathbf{M}}^{(s)}$ and $\hat{\mathbf{K}}^{(s)}$ are substructure system matrices representing mass and stiffness, with internal dynamics truncated using a set of dominant modes $N_{i,d}$, and $\hat{\mathbf{F}}$ is the corresponding force vector. Referred to as the Craig-Bampton method in the literature [1], this approach is characterised by coupled mass and decoupled stiffness between internal and interface regions, as evident in Eq. (9). Considering a primal coupling approach introduced in Eqs. (3) and (5), the reduced global equations of motion in Eq. (9) can be derived as:

$$\hat{\mathbf{M}}_{\mathbf{g}} \hat{\mathbf{u}}_{\mathbf{g}} + \hat{\mathbf{K}}_{\mathbf{g}} \hat{\mathbf{u}}_{\mathbf{g}} = \hat{\mathbf{F}}_{\mathbf{g}}, \quad (10)$$

where $\hat{\mathbf{M}}_{\mathbf{g}}$ and $\hat{\mathbf{K}}_{\mathbf{g}}$ are respectively global mass and stiffness matrices with truncated internal dynamics using CB method, and $\hat{\mathbf{F}}_{\mathbf{g}}$ is the corresponding global force vector. The size of the system matrices and force vector are respectively $\hat{N} \times \hat{N}$ and $\hat{N} \times 1$, with $\hat{N} = N_{i,d} + N_b$.

In the original CB method, the residual substructural modes are truncated without any consideration. To acquire an improved accuracy of the reduced matrices, Kim and Lee [19] proposed an ECB method where the residual substructural modes are in addition to the dominant structural modes included in the transformation matrix. Considering the ECB approach in [19], the substructure displacement vector can be approximated as:

$$\mathbf{u}^{(s)} \approx \bar{\mathbf{T}}_{\text{ECB}}^{(s)} \bar{\mathbf{u}}^{(s)} = \bar{\mathbf{T}}_{\text{ECB}}^{(s)} \begin{Bmatrix} \mathbf{q}_{i,d}^{(s)} \\ \mathbf{u}_b^{(s)} \end{Bmatrix}, \quad \bar{\mathbf{T}}_{\text{ECB}} = \bar{\mathbf{T}}_{\text{CB}}^{(s)} + \bar{\mathbf{T}}_{\text{R}}^{(s)}, \quad (11a)$$

$$\bar{\mathbf{T}}_{\text{R}}^{(s)} = \begin{bmatrix} \mathbf{0} & \mathbf{F}_{\text{rs}}^{(s)} \left[-\mathbf{M}_{ii}^{(s)} \mathbf{K}_{ii}^{(s)-1} \mathbf{K}_{ib}^{(s)} + \mathbf{M}_{ib}^{(s)} \right] \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \bar{\mathbf{M}}_{\text{CB}}^{(s)-1} \bar{\mathbf{K}}_{\text{CB}}^{(s)}, \quad (11b)$$

$$\mathbf{F}_{\text{rs}}^{(s)} = \mathbf{K}_{ii}^{(s)} - \Phi_{i,d}^{(s)} \Lambda_{i,d}^{(s)-1} \Phi_{i,d}^{(s)\top}, \quad \Lambda_{i,d}^{(s)} = \Phi_{i,d}^{(s)\top} \mathbf{K}_{ii}^{(s)} \Phi_{i,d}^{(s)}. \quad (11c)$$

The equation above indicates that the CB transformation matrix $\bar{\mathbf{T}}_{\text{CB}}^{(s)}$ has been supplemented with another transformation matrix, namely $\bar{\mathbf{T}}_{\text{R}}^{(s)}$ which contains residual modal effect $\mathbf{F}_{\text{rs}}^{(s)}$. With the ECB transformation matrix derived in Eq. (11), the substructure equations of motion in Eq. (1) can be approximated as follows:

$$\bar{\mathbf{T}}_{\text{ECB}}^{(s)\top} \mathbf{M}^{(s)} \bar{\mathbf{T}}_{\text{ECB}}^{(s)} \begin{Bmatrix} \ddot{\mathbf{q}}_{i,d}^{(s)} \\ \ddot{\mathbf{u}}_b^{(s)} \end{Bmatrix} + \bar{\mathbf{T}}_{\text{ECB}}^{(s)\top} \mathbf{K}^{(s)} \bar{\mathbf{T}}_{\text{ECB}}^{(s)} \begin{Bmatrix} \mathbf{q}_{i,d}^{(s)} \\ \mathbf{u}_b^{(s)} \end{Bmatrix} = \bar{\mathbf{T}}_{\text{ECB}}^{(s)\top} \begin{Bmatrix} \mathbf{f}_i^{(s)} \\ \mathbf{f}_b^{(s)} \end{Bmatrix}, \quad (12)$$

with

$$\bar{\mathbf{M}}^{(s)} = \bar{\mathbf{T}}_{\text{ECB}}^{(s)\top} \mathbf{M}^{(s)} \bar{\mathbf{T}}_{\text{ECB}}^{(s)} = \begin{bmatrix} \bar{\mathbf{M}}_{ii}^{(s)} & \bar{\mathbf{M}}_{ib}^{(s)} \\ \bar{\mathbf{M}}_{bi}^{(s)} & \bar{\mathbf{M}}_{bb}^{(s)} \end{bmatrix}, \quad (13a)$$

$$\bar{\mathbf{K}}^{(s)} = \bar{\mathbf{T}}_{\text{ECB}}^{(s)\top} \mathbf{K}^{(s)} \bar{\mathbf{T}}_{\text{ECB}}^{(s)} = \begin{bmatrix} \bar{\mathbf{K}}_{ii}^{(s)} & \bar{\mathbf{K}}_{ib}^{(s)} \\ \bar{\mathbf{K}}_{bi}^{(s)} & \bar{\mathbf{K}}_{bb}^{(s)} \end{bmatrix}, \quad (13b)$$

$$\bar{\mathbf{f}}^{(s)} = \bar{\mathbf{T}}_{\text{ECB}}^{(s)\top} \begin{Bmatrix} \mathbf{f}_i^{(s)} \\ \mathbf{f}_b^{(s)} \end{Bmatrix} = \begin{Bmatrix} \bar{\mathbf{f}}_i^{(s)} \\ \bar{\mathbf{f}}_b^{(s)} \end{Bmatrix}, \quad (13c)$$

where $\bar{\mathbf{M}}^{(s)}$ and $\bar{\mathbf{K}}^{(s)}$ are system matrices for which the internal DOFs are truncated using a fixed-interface based ECB method. In Eq. (13), it is evident that the off-diagonal terms of $\bar{\mathbf{K}}^{(s)}$ are non-zero. This is in contrast to the classical CB method, which features decoupled internal and interface dynamics, as demonstrated in Eq. (9). As a result, the ECB method exhibits stiffness coupling between internal and boundary regions. For further insight into CMS methods, the reader is referred to [7,11]. With the primal coupling approach in Eqs. (3) and (5) the reduced system in Eq. (13) can be assembled to derive the reduced equations of motion as follows:

$$\bar{\mathbf{M}}_{\mathbf{g}} \ddot{\mathbf{u}}_{\mathbf{g}} + \bar{\mathbf{K}}_{\mathbf{g}} \mathbf{u}_{\mathbf{g}} = \bar{\mathbf{F}}_{\mathbf{g}} \quad (14)$$

where $\bar{\mathbf{M}}_{\mathbf{g}}$, $\bar{\mathbf{K}}_{\mathbf{g}}$ and $\bar{\mathbf{F}}_{\mathbf{g}}$ are respectively global mass, stiffness matrices, and force vector, with truncated internal dynamics using ECB method. The substructures are coupled at their boundaries with constraint modes. The number of dominant internal modes retained and included in the reduction for substructure s was defined as $N_{i,d}^{(s)}$ in Eq. (6). Hence, the total number of retained modes associated with all substructures must become $N_{i,d} = \sum_{k=1}^{N_s} N_{i,d}^{(s)}$. The number of DOFs in the reduced system becomes $\bar{N} = N_{i,d} + N_b$, which is fewer than the total DOFs before reduction, denoted as N . Note that the total internal DOFs were originally N_i and the total number of the internal DOFs after reduction has become $N_{i,d} \ll N_i$.

The equations presented above may still comprise a significant number of DOFs, depending on the extent of coupling between the substructures at their interfaces. As such, in the following section, techniques for reducing the size of the interface region will be described.

3. Reduction of interface DOFs

In the previous section, the internal dynamics of each substructure were truncated using a set of fixed-interface normal modes and subsequently coupled at their interfaces using constraint modes. In other words, the substructure internal DOFs were reduced whilst the physical interface DOFs were retained. Since the motion on the substructural interface is described by the constraint modes, the number of DOFs associated with the constraint modes may result in a further computational cost. In particular, for a sufficiently fine FE discretisation with many substructural components, the size of the CMS model may be dominated by the constraint mode DOFs. Therefore, a reduced interface approach can be considered to reduce computational burden and achieve a reduced set of system matrices. In this section, two interface reduction techniques will be introduced to truncate the interface dynamics of system matrices with truncated internal dynamics. The first option, introduced in Section 3.1, is based on the classical approach proposed by Castanier et al. [30] to truncate interface dynamics of the CB method. This approach is referred to as IR1 in subsequent derivations. The second option is a refinement of Castanier et al.'s [30] approach, which accounts for the coupled stiffness terms between internal and interface DOFs. This refinement is particularly important because, in contrast to the CB method, the ECB method comprises coupled internal and interface stiffness. This refined approach is denoted as IR2 in the following derivations in Section 3.2.

To reduce the interface DOFs, a similar approach as for the internal DOFs reduction can be considered, namely truncating the interface dynamics with a set of fixed-internal modes obtained from computing an eigenvalue analysis for the interface DOFs as:

$$\left(\mathbf{K}_{bb} - \omega_{b,j}^2 \mathbf{M}_{bb}\right) \boldsymbol{\phi}_{b,j} = \mathbf{0}, \quad j = 1, 2, \dots, N_{b,d}, \quad (15)$$

where $\omega_{b,j}^2$ and $\boldsymbol{\phi}_{b,j}$ are j th eigenvalue and eigenvector associated with the interface DOFs, respectively. The interface mode $\boldsymbol{\phi}_{b,j}$ is often referred to as system-level Characteristic Constraint (CC) mode. The number of dominant modes associated with the interface boundary is labelled as $N_{b,d}$, which is less than the total number of boundary modes N_b , i.e. $N_{b,d} \ll N_b$. In this study, the number of dominant interface modes, denoted as $N_{b,d}$, is selected based on mode convergence, as shown in Figs. 9, 4, and 17. The entry system matrices in Eq. (15) are the interface partitions of the assembled (global) system matrices defined in Eq. (10) for CB and Eq. (14) for ECB, respectively.

3.1. Interface reduction technique

Selecting a number of dominant interface modes, the structural displacement vector of the boundary DOFs can be approximated as:

$$\mathbf{u}_b \approx \boldsymbol{\Phi}_{b,d} \mathbf{q}_{b,d}, \quad (16)$$

where the modal matrix $\boldsymbol{\Phi}_{b,d}$ contains dominant interface eigenvectors and has the dimension $N_b \times N_{b,d}$, where $N_{b,d} \ll N_b$, and $\mathbf{q}_{b,d}$ is the eigenvector's associated generalised coordinates. Truncating the boundary DOFs using a set of dominant modes, the displacement vector of the coupled systems in Eqs. (10) and (14) can be further reduced as follows:

$$\hat{\mathbf{u}}_g \approx \bar{\mathbf{T}}_{\text{IR1}} \begin{Bmatrix} \mathbf{q}_{i,d} \\ \mathbf{q}_{b,d} \end{Bmatrix}, \quad \bar{\mathbf{u}}_g \approx \bar{\mathbf{T}}_{\text{IR1}} \begin{Bmatrix} \mathbf{q}_{i,d} \\ \mathbf{q}_{b,d} \end{Bmatrix}, \quad \bar{\mathbf{T}}_{\text{IR1}} = \begin{bmatrix} \mathbf{I}_{i,d} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Phi}_{b,d} \end{bmatrix}, \quad (17)$$

where $\hat{\mathbf{u}}_g$ and $\bar{\mathbf{u}}_g$ are displacement vectors associated with the equations of motion, for which the internal dynamics are truncated using CB and ECB methods, respectively. The interface transformation matrix $\bar{\mathbf{T}}_{\text{IR1}}$ was originally proposed by [30] for use in the interface reduction of the original Craig-Bampton method. The displacement vector, after truncation of the interface DOFs, now contains generalised internal and interface coordinates. As evident in Eq. (17), $\bar{\mathbf{T}}_{\text{IR1}}$ implies decoupled internal and interface dynamics, as indicated by the zeros in the off-diagonal terms. This approach aligns with the decoupled stiffness characteristics of CB, as demonstrated in Eq. (9). However, the coupled stiffness characteristics of ECB in Eq. (13) make this approach unsuitable for ECB, as shown further through the application examples in Section 4. Considering the $\bar{\mathbf{T}}_{\text{IR1}}$ to truncate interface dynamics of the global system matrices for which internal dynamics is truncated using the CB method in Eq. (9), the equations of motion in Eq. (10) can be further reduced to

$$\bar{\mathbf{T}}_{\text{IR1}}^T \hat{\mathbf{M}}_g \bar{\mathbf{T}}_{\text{IR1}} \begin{Bmatrix} \ddot{\mathbf{q}}_{i,d} \\ \ddot{\mathbf{q}}_{b,d} \end{Bmatrix} + \bar{\mathbf{T}}_{\text{IR1}}^T \hat{\mathbf{K}}_g \bar{\mathbf{T}}_{\text{IR1}} \begin{Bmatrix} \mathbf{q}_{i,d} \\ \mathbf{q}_{b,d} \end{Bmatrix} = \bar{\mathbf{T}}_{\text{IR1}}^T \begin{Bmatrix} \hat{\mathbf{f}}_i \\ \hat{\mathbf{f}}_b \end{Bmatrix}, \quad (18)$$

with

$$\check{\mathbf{M}} = \bar{\mathbf{T}}_{\text{IR1}}^T \hat{\mathbf{M}}_g \bar{\mathbf{T}}_{\text{IR1}} = \begin{bmatrix} \check{\mathbf{M}}_{ii} & \check{\mathbf{M}}_{ib} \\ \check{\mathbf{M}}_{bi} & \check{\mathbf{M}}_{bb} \end{bmatrix}, \quad (19a)$$

$$\check{\mathbf{K}} = \bar{\mathbf{T}}_{\text{IR1}}^T \hat{\mathbf{K}}_g \bar{\mathbf{T}}_{\text{IR1}} = \begin{bmatrix} \check{\mathbf{K}}_{ii} & \check{\mathbf{K}}_{ib} \\ \check{\mathbf{K}}_{bi} & \check{\mathbf{K}}_{bb} \end{bmatrix}, \quad (19b)$$

$$\check{\mathbf{f}} = \bar{\mathbf{T}}_{\text{IR1}}^T \begin{Bmatrix} \hat{\mathbf{f}}_i \\ \hat{\mathbf{f}}_b \end{Bmatrix} = \begin{Bmatrix} \check{\mathbf{f}}_i \\ \check{\mathbf{f}}_b \end{Bmatrix}. \quad (19c)$$

The component terms of the system matrices in Eq. (19) are computed using the components of the system matrices in Eq. (10), as:

$$\tilde{\mathbf{M}}_{ii} = \hat{\mathbf{M}}_{ii}, \quad \tilde{\mathbf{M}}_{ib} = \hat{\mathbf{M}}_{ii} \Phi_{b,d}, \quad \tilde{\mathbf{M}}_{bi} = \tilde{\mathbf{M}}_{ib}^T, \quad \tilde{\mathbf{M}}_{bb} = \Phi_{b,d}^T \hat{\mathbf{M}}_{bb} \Phi_{b,d}, \quad (20a)$$

$$\tilde{\mathbf{K}}_{ii} = \bar{\mathbf{K}}_{ii}, \quad \tilde{\mathbf{K}}_{ib} = \hat{\mathbf{K}}_{ii} \Phi_{b,d}, \quad \tilde{\mathbf{K}}_{bi} = \tilde{\mathbf{K}}_{ib}^T, \quad \tilde{\mathbf{K}}_{bb} = \Phi_{b,d}^T \hat{\mathbf{K}}_{bb} \Phi_{b,d}, \quad (20b)$$

$$\tilde{\mathbf{f}}_i = \hat{\mathbf{f}}_i, \quad \tilde{\mathbf{f}}_b = \Phi_{b,d}^T \hat{\mathbf{f}}_b, \quad (20c)$$

where it is noted that the interface partition of the CB system matrices, i.e. $\hat{\mathbf{M}}_{bb}$ and $\hat{\mathbf{K}}_{bb}$ are reduced using classical modal truncation. However, the truncated internal partition of the CB system matrices, i.e. $\hat{\mathbf{M}}_{ii}$ and $\hat{\mathbf{K}}_{ii}$ are just retained. Accordingly, the equations of motion in Eq. (10) can be further reduced to

$$\tilde{\mathbf{M}} \ddot{\mathbf{u}} + \tilde{\mathbf{K}} \dot{\mathbf{u}} = \tilde{\mathbf{F}}, \quad (21)$$

where the system matrices $\tilde{\mathbf{M}}$, $\tilde{\mathbf{K}}$ and force vector $\tilde{\mathbf{F}}$ are those for which the internal and boundary DOFs are reduced using the CB method and interface reduction matrix $\tilde{\mathbf{T}}_{IR1}$, respectively. Hence the size of the mentioned system matrices and the force vector are respectively $\tilde{N} \times \tilde{N}$ and $\tilde{N} \times 1$, where $\tilde{N} = N_{i,d} + N_{b,d}$, which is smaller than $\hat{N}$, $\bar{N}$ and N . This combination of CB and IR1 is denoted CB-IR1 in the application examples in Section 4.

Considering $\tilde{\mathbf{T}}_{IR1}$ to truncate interface dynamics of the global system matrices for which the internal dynamics are truncated using the ECB method, the system matrices in Eq. (14) can be further reduced to

$$\tilde{\mathbf{T}}_{IR1}^T \tilde{\mathbf{M}}_g \tilde{\mathbf{T}}_{IR1} \begin{Bmatrix} \ddot{\mathbf{q}}_{i,d} \\ \ddot{\mathbf{q}}_{b,d} \end{Bmatrix} + \tilde{\mathbf{T}}_{IR1}^T \tilde{\mathbf{K}}_g \tilde{\mathbf{T}}_{IR1} \begin{Bmatrix} \mathbf{q}_{i,d} \\ \mathbf{q}_{b,d} \end{Bmatrix} = \tilde{\mathbf{T}}_{IR1}^T \begin{Bmatrix} \tilde{\mathbf{f}}_i \\ \tilde{\mathbf{f}}_b \end{Bmatrix}, \quad (22)$$

with

$$\tilde{\mathbf{M}} = \tilde{\mathbf{T}}_{IR1}^T \tilde{\mathbf{M}}_g \tilde{\mathbf{T}}_{IR1} = \begin{bmatrix} \tilde{\mathbf{M}}_{ii} & \tilde{\mathbf{M}}_{ib} \\ \tilde{\mathbf{M}}_{bi} & \tilde{\mathbf{M}}_{bb} \end{bmatrix}, \quad (23a)$$

$$\tilde{\mathbf{K}} = \tilde{\mathbf{T}}_{IR1}^T \tilde{\mathbf{K}}_g \tilde{\mathbf{T}}_{IR1} = \begin{bmatrix} \tilde{\mathbf{K}}_{ii} & \tilde{\mathbf{K}}_{ib} \\ \tilde{\mathbf{K}}_{bi} & \tilde{\mathbf{K}}_{bb} \end{bmatrix}, \quad (23b)$$

$$\tilde{\mathbf{F}} = \tilde{\mathbf{T}}_{IR1}^T \begin{Bmatrix} \tilde{\mathbf{f}}_i \\ \tilde{\mathbf{f}}_b \end{Bmatrix} = \begin{Bmatrix} \tilde{\mathbf{f}}_i \\ \tilde{\mathbf{f}}_b \end{Bmatrix}, \quad (23c)$$

where the elements of the system matrices $\tilde{\mathbf{M}}$ and $\tilde{\mathbf{K}}$, and the force vector $\tilde{\mathbf{F}}$ are defined as:

$$\tilde{\mathbf{M}}_{ii} = \bar{\mathbf{M}}_{ii}, \quad \tilde{\mathbf{M}}_{ib} = \bar{\mathbf{M}}_{ii} \Phi_{b,d}, \quad \tilde{\mathbf{M}}_{bi} = \tilde{\mathbf{M}}_{ib}^T, \quad \tilde{\mathbf{M}}_{bb} = \Phi_{b,d}^T \bar{\mathbf{M}}_{bb} \Phi_{b,d}, \quad (24a)$$

$$\tilde{\mathbf{K}}_{ii} = \bar{\mathbf{K}}_{ii}, \quad \tilde{\mathbf{K}}_{ib} = \bar{\mathbf{K}}_{ii} \Phi_{b,d}, \quad \tilde{\mathbf{K}}_{bi} = \tilde{\mathbf{K}}_{ib}^T, \quad \tilde{\mathbf{K}}_{bb} = \Phi_{b,d}^T \bar{\mathbf{K}}_{bb} \Phi_{b,d}. \quad (24b)$$

$$\tilde{\mathbf{f}}_i = \bar{\mathbf{f}}_i, \quad \tilde{\mathbf{f}}_b = \Phi_{b,d}^T \bar{\mathbf{f}}_b. \quad (24c)$$

The equation above indicates that the interface portions of the system matrices, i.e., $\tilde{\mathbf{M}}_{bb}$ and $\tilde{\mathbf{K}}_{bb}$, and the force vector $\tilde{\mathbf{f}}_b$, are truncated using classical modal truncation. However, this approach does not account for the stiffness coupling present in the ECB method, as shown in Eq. (13). With the system matrices and force vector in Eq. (23), the equations of motion in Eq. (14) can be reduced to

$$\tilde{\mathbf{M}} \ddot{\mathbf{u}} + \tilde{\mathbf{K}} \dot{\mathbf{u}} = \tilde{\mathbf{F}}, \quad (25)$$

where the sizes of the system matrices $\tilde{\mathbf{M}}$ and $\tilde{\mathbf{K}}$, and the force vector $\tilde{\mathbf{F}}$ are $\tilde{N} \times \tilde{N}$ and $\tilde{N} \times 1$, respectively, where $\tilde{N} = N_{i,d} + N_{b,d}$. This size is smaller than that of $\hat{N}$, $\bar{N}$, and N . The approach using a combination of ECB and IR1 is denoted as ECB-IR1 in the application examples in Section 4.

3.2. Refined interface reduction technique

Referring back to Eq. (13), it is evident that the off-diagonal terms of ECB, which indicates the interaction between the truncated internal and boundary DOFs, are not entirely decoupled. These off-diagonal terms serve as a forcing function, resulting in the excitation of the internal dynamics by the motion at the interface. Consequently, to project the truncated internal dynamics onto the interface, the consideration of an additional constraint mode matrix is necessary to facilitate the transfer of the vibration energy. In this section, the interface reduction matrix presented in the previous section denoted as $\tilde{\mathbf{T}}_{IR1}$, will be modified to enhance its accuracy. This modification is intended for cases where internal dynamics are truncated using the ECB method in Eq. (14).

Considering an additional coupling term, the interface boundary displacement vector in Eq. (16) can be approximated as:

$$\mathbf{u}_b \approx \Xi \mathbf{q}_{i,d} + \Phi_{b,d} \mathbf{q}_{b,d}, \quad (26)$$

where Ξ is a constraint mode matrix which projects the internal DOFs onto the interface DOFs, $\Phi_{b,d}$ is a mode matrix containing dominant interface modes, obtained in Eq. (15) with a column size of $N_{b,d}$, and $\mathbf{q}_{b,d}$ is its associated generalised coordinates. The shape of the constraint coupling term Ξ is similar to a static condensation of the interface DOFs. This consideration is valid for low-frequency vibrations considering the interface DOFs respond quasi-statically to the truncated internal DOFs.

The global displacement vector in Eq. (14) can be further reduced to

$$\tilde{\mathbf{u}}_g \approx \tilde{\mathbf{T}}_{\text{IR2}} \begin{Bmatrix} \mathbf{q}_{i,d} \\ \mathbf{q}_{b,d} \end{Bmatrix}, \quad \tilde{\mathbf{u}}_g = \begin{Bmatrix} \mathbf{q}_{i,d} \\ \mathbf{q}_{b,d} \end{Bmatrix}, \quad \tilde{\mathbf{T}}_{\text{IR2}} = \begin{bmatrix} \mathbf{I}_{i,d} & \mathbf{0} \\ \Xi & \Phi_{b,d} \end{bmatrix}, \quad \Xi = -\tilde{\mathbf{K}}_{bb}^{-1} \tilde{\mathbf{K}}_{bi}, \quad (27)$$

where Ξ is computed using components of $\tilde{\mathbf{K}}_g$ Eq. (14), for which the internal DOFs are reduced using the ECB method and has the dimension of $\tilde{N}_b \times \tilde{N}_{i,d}$. It is noted that the transformation matrix $\tilde{\mathbf{T}}_{\text{IR2}}$ in Eq. (27) is similar to the $\tilde{\mathbf{T}}_{\text{IR1}}$ in Eq. (17), however, with an additional coupling term Ξ .

Considering the refined interface reduction technique which takes the coupling between the truncated internal and interface DOFs into account, i.e. $\tilde{\mathbf{T}}_{\text{IR2}}$, the equations of motion in Eq. (14) can be further reduced to

$$\tilde{\mathbf{T}}_{\text{IR2}}^T \tilde{\mathbf{M}}_g \tilde{\mathbf{T}}_{\text{IR2}} \begin{Bmatrix} \ddot{\mathbf{q}}_{i,d} \\ \ddot{\mathbf{q}}_{b,d} \end{Bmatrix} + \tilde{\mathbf{T}}_{\text{IR2}}^T \tilde{\mathbf{K}}_g \tilde{\mathbf{T}}_{\text{IR2}} \begin{Bmatrix} \mathbf{q}_{i,d} \\ \mathbf{q}_{b,d} \end{Bmatrix} = \tilde{\mathbf{T}}_{\text{IR2}}^T \begin{Bmatrix} \tilde{\mathbf{f}}_i \\ \tilde{\mathbf{f}}_b \end{Bmatrix}, \quad (28)$$

where

$$\tilde{\mathbf{M}} = \tilde{\mathbf{T}}_{\text{IR2}}^T \tilde{\mathbf{M}}_g \tilde{\mathbf{T}}_{\text{IR2}} = \begin{bmatrix} \tilde{\mathbf{M}}_{ii} & \tilde{\mathbf{M}}_{ib} \\ \tilde{\mathbf{M}}_{bi} & \tilde{\mathbf{M}}_{bb} \end{bmatrix}, \quad (29a)$$

$$\tilde{\mathbf{K}} = \tilde{\mathbf{T}}_{\text{IR2}}^T \tilde{\mathbf{K}}_g \tilde{\mathbf{T}}_{\text{IR2}} = \begin{bmatrix} \tilde{\mathbf{K}}_{ii} & \tilde{\mathbf{K}}_{ib} \\ \tilde{\mathbf{K}}_{bi} & \tilde{\mathbf{K}}_{bb} \end{bmatrix}, \quad (29b)$$

$$\tilde{\mathbf{F}} = \tilde{\mathbf{T}}_{\text{IR2}}^T \begin{Bmatrix} \tilde{\mathbf{f}}_i \\ \tilde{\mathbf{f}}_b \end{Bmatrix} = \begin{Bmatrix} \tilde{\mathbf{f}}_i \\ \tilde{\mathbf{f}}_b \end{Bmatrix}, \quad (29c)$$

with

$$\tilde{\mathbf{M}}_{ii} = \tilde{\mathbf{M}}_{ii} + \Xi^T \tilde{\mathbf{M}}_{bi} + \tilde{\mathbf{M}}_{bi} \Xi + \Xi^T \tilde{\mathbf{M}}_{bb} \Xi, \quad (30a)$$

$$\tilde{\mathbf{M}}_{ib} = \tilde{\mathbf{M}}_{ib} \Phi_{b,d} + \Xi^T \tilde{\mathbf{M}}_{bb} \Phi_{b,d}, \quad \tilde{\mathbf{M}}_{bi} = \tilde{\mathbf{M}}_{ib}^T, \quad (30b)$$

$$\tilde{\mathbf{M}}_{bb} = \Phi_{b,d}^T \tilde{\mathbf{M}}_{bb} \Phi_{b,d}, \quad (30c)$$

$$\tilde{\mathbf{K}}_{ii} = \tilde{\mathbf{K}}_{ii} + \Xi^T \tilde{\mathbf{K}}_{bi} + \tilde{\mathbf{K}}_{bi} \Xi + \Xi^T \tilde{\mathbf{K}}_{bb} \Xi, \quad (30d)$$

$$\tilde{\mathbf{K}}_{ib} = \tilde{\mathbf{K}}_{ib} \Phi_{b,d} + \Xi^T \tilde{\mathbf{K}}_{bb} \Phi_{b,d}, \quad \tilde{\mathbf{K}}_{bi} = \tilde{\mathbf{K}}_{ib}^T, \quad (30e)$$

$$\tilde{\mathbf{K}}_{bb} = \Phi_{b,d}^T \tilde{\mathbf{K}}_{bb} \Phi_{b,d}, \quad (30f)$$

$$\tilde{\mathbf{f}}_i = \tilde{\mathbf{f}}_i, \quad \tilde{\mathbf{f}}_b = \Xi^T \tilde{\mathbf{f}}_i + \Phi_{b,d}^T \tilde{\mathbf{f}}_b, \quad (30g)$$

where the system matrices $\tilde{\mathbf{M}}$ and $\tilde{\mathbf{K}}$ and force vector $\tilde{\mathbf{F}}$ are those for which the internal and boundary DOFs are reduced using ECB method and the refined interface reduction transformation matrix $\tilde{\mathbf{T}}_{\text{IR2}}$, respectively. The components of the reduced system matrices in Eq. (30) clearly show that the truncated internal and coupling terms, i.e. $\tilde{\mathbf{M}}_{ii}$ and $\tilde{\mathbf{M}}_{ib} = \tilde{\mathbf{M}}_{bi}^T$ have been enhanced by additional terms accounting for the coupling between the truncated internal and interface DOFs, cf. Eqs. (20) and (24). Accordingly, the equations of motion in Eq. (14) can be further reduced to

$$\tilde{\mathbf{M}} \ddot{\tilde{\mathbf{u}}} + \tilde{\mathbf{K}} \tilde{\mathbf{u}} = \tilde{\mathbf{f}}, \quad (31)$$

where the size of the system matrices $\tilde{\mathbf{M}}$ and $\tilde{\mathbf{K}}$ is $\tilde{N} \times \tilde{N}$ and the size of the force vector $\tilde{\mathbf{f}}$ is $\tilde{N} \times 1$ with $\tilde{N} = N_{i,d} + N_{b,d}$, which is smaller than both $\tilde{N}$ and N , Table 1. The approach using a combination of ECB and IR2 is denoted ECB-IR2 in the application examples in Section 4.

With the aim of providing a concise summary of each interface reduction technique and its key attributes, a brief comparison is provided in Table 1. Additionally, to assist the implementation of the interface reduction techniques, a supplementary step-by-step procedure is provided in Algorithm 1.

Table 1

Comparison between CB with interface reduction (CB-IR1), ECB with interface reduction (ECB-IR1), and ECB with refined interface reduction (ECB-IR2). The checkmark denotes valid operations.

	CB-IR1	ECB-IR1	ECB-IR2
Transformation matrix	$\tilde{\mathbf{T}}_{\text{IR1}} \tilde{\mathbf{T}}_{\text{CB}}$	$\tilde{\mathbf{T}}_{\text{IR1}} \tilde{\mathbf{T}}_{\text{ECB}}$	$\tilde{\mathbf{T}}_{\text{IR2}} \tilde{\mathbf{T}}_{\text{ECB}}$
Reduced mass matrix	$\tilde{\mathbf{T}}_{\text{IR1}}^T \tilde{\mathbf{T}}_{\text{CB}}^T \mathbf{M} \tilde{\mathbf{T}}_{\text{CB}} \tilde{\mathbf{T}}_{\text{IR1}}$	$\tilde{\mathbf{T}}_{\text{IR1}}^T \tilde{\mathbf{T}}_{\text{ECB}}^T \mathbf{M} \tilde{\mathbf{T}}_{\text{ECB}} \tilde{\mathbf{T}}_{\text{IR1}}$	$\tilde{\mathbf{T}}_{\text{IR2}}^T \tilde{\mathbf{T}}_{\text{ECB}}^T \mathbf{M} \tilde{\mathbf{T}}_{\text{ECB}} \tilde{\mathbf{T}}_{\text{IR2}}$
Reduced stiffness matrix	$\tilde{\mathbf{T}}_{\text{IR1}}^T \tilde{\mathbf{T}}_{\text{CB}}^T \mathbf{K} \tilde{\mathbf{T}}_{\text{CB}} \tilde{\mathbf{T}}_{\text{IR1}}$	$\tilde{\mathbf{T}}_{\text{IR1}}^T \tilde{\mathbf{T}}_{\text{ECB}}^T \mathbf{K} \tilde{\mathbf{T}}_{\text{ECB}} \tilde{\mathbf{T}}_{\text{IR1}}$	$\tilde{\mathbf{T}}_{\text{IR2}}^T \tilde{\mathbf{T}}_{\text{ECB}}^T \mathbf{K} \tilde{\mathbf{T}}_{\text{ECB}} \tilde{\mathbf{T}}_{\text{IR2}}$
Size of the reduced matrices	$\tilde{N} = N_{i,d} + N_{b,d}$	$\tilde{N} = N_{i,d} + N_{b,d}$	$\tilde{N} = N_{i,d} + N_{b,d}$
Reduced internal dynamics	✓	✓	✓
Reduced interface dynamics	✓	✓	✓

Algorithm 1 Implementation of interface reduction using CB-IR1, ECB-IR1, and ECB-IR2.

Input: Full model substructure system matrices $(\mathbf{M}^{(s)}, \mathbf{K}^{(s)}, \mathbf{f}^{(s)})$

Output: Reduced system matrices $((\tilde{\mathbf{M}}, \tilde{\mathbf{K}}, \tilde{\mathbf{f}}), (\tilde{\mathbf{M}}, \tilde{\mathbf{K}}, \tilde{\mathbf{f}}), (\tilde{\mathbf{M}}, \tilde{\mathbf{K}}, \tilde{\mathbf{f}}))$

procedure

1. Solve the internal eigenvalue problem for substructure $s = 1, 2, \dots, N_s$
 $(\mathbf{K}_{ii}^{(s)} - \omega_{i,j}^2 \mathbf{M}_{ii}^{(s)}) \boldsymbol{\phi}_{i,j}^{(s)} = \mathbf{0}, \quad j = 1, 2, \dots, N_{i,d}^{(s)}, \quad \text{in Eq. (6)}$
2. Compute the CB reduction matrix $\tilde{\mathbf{T}}_{\text{CB}}^{(s)}$, in Eq. (7)
3. Compute the ECB reduction matrix $\tilde{\mathbf{T}}_{\text{ECB}}^{(s)}$, in Eq. (11)
4. Truncate the interior dynamics of each substructure and compute
 $\tilde{\mathbf{M}}_g, \tilde{\mathbf{K}}_g, \tilde{\mathbf{f}}_g$ and $\tilde{\mathbf{M}}_g, \tilde{\mathbf{K}}_g, \tilde{\mathbf{f}}_g$, in Eqs. (10) and (14)
5. Solve the system-level interface eigenvalue problem and obtain CC modes:
 $(\mathbf{K}_{bb} - \omega_{b,j}^2 \mathbf{M}_{bb}) \boldsymbol{\phi}_{b,j} = \mathbf{0}, \quad j = 1, 2, \dots, N_{b,d}, \quad \text{in Eq. (15)}$
6. Compute the interface reduction transformation $\tilde{\mathbf{T}}_{\text{IR1}}$ in Eq. (17)
7. Compute the refined interface reduction transformation matrix $\tilde{\mathbf{T}}_{\text{IR2}}$ in Eq. (27)
8. Truncate system-level interface dynamics and compute:
 - CB-IR1 system matrices and force vector: $\tilde{\mathbf{M}}, \tilde{\mathbf{K}}, \tilde{\mathbf{f}}$
 - ECB-IR1 system matrices and force vector: $\tilde{\mathbf{M}}, \tilde{\mathbf{K}}, \tilde{\mathbf{f}}$
 - ECB-IR2 system matrices and force vector: $\tilde{\mathbf{M}}, \tilde{\mathbf{K}}, \tilde{\mathbf{f}}$

end procedure

4. Application examples

In this section, the interface reduction techniques discussed in the previous section are applied to reduce the interface DOFs in three examples. To assess each model's performance, the results are compared with those from the full model, CB model, and ECB model. The accuracy of each model is then evaluated by considering the relative frequency error as:

$$\text{Relative frequency error} = \frac{|\bar{f}_i - f_i|}{f_i}, \quad (32)$$

where f_i represents the resonance frequency of component i for comparison. An over-bar, as previously explained, indicates an approximate quantity. Therefore, $\bar{f}_i$ and f_i respectively correspond to the frequency components of the reduced and full models.

The accuracy of the models is further evaluated in the frequency domain by considering their harmonic response due to the application of a harmonic unit load. The equations of motion in the frequency domain in the absence of damping can be expressed as:

$$\mathbf{Z}(\omega) \mathbf{u}(\omega) = \mathbf{F}(\omega), \quad \mathbf{Z}(\omega) \stackrel{\Delta}{=} -\omega^2 \mathbf{M} + \mathbf{K}, \quad (33)$$

where $\mathbf{Z}(\omega)$ represents a block-diagonal matrix containing the dynamic stiffness of the coupled substructures. The system matrices $\mathbf{M}$ and $\mathbf{K}$ are mass and stiffness matrices, respectively. The response vector is denoted by $\mathbf{u}(\omega)$ and the force amplitude vector is defined as $\mathbf{F}(\omega)$. The response is a function of the driving angular frequency ω in [rad/s]. Hence, the dynamic stiffness matrix $\mathbf{Z}(\omega)$ as well as the force vector $\mathbf{F}(\omega)$ are also functions of the driving frequency ω . The transfer function representing normalised displacement response under unit loading is often referred to as Admittance. For additional insight into the admittance response, the representation plots are supplemented with the phase shift angle of the admittance at each discrete frequency step.

To evaluate the accuracy of the vibration modes associated with the reduced models in comparison with the full model, the statistical indicator Modal Assurance Criterion (MAC) can be employed, [40,41]. The advantages of employing MAC stem from the fact that it is a quantitative tool for comparing the degree of consistency between two modes, whilst being sensitive to large

Table 2
Number of internal and interface modes computed for the stiffened plate example in Section 4.1.

Method	Number of internal modes retained						Number of interface modes retained	Total number of DOFs
	$N_{i,d}^{(1)}$	$N_{i,d}^{(2)}$	$N_{i,d}^{(3)}$	$N_{i,d}^{(4)}$	$N_{i,d}^{(5)}$	$N_{i,d}^{(6)}$		
Full model	4542	3252	1584	1584	2214	2214	972	$N = 16362$
CB	20	10	5	5	10	10	972	$\tilde{N} = 1032$
ECB	20	10	5	5	10	10	972	$\tilde{N} = 1032$
CB-IR1	20	10	5	5	10	10	196	$\tilde{N} = 256$
ECB-IR1	20	10	5	5	10	10	196	$\tilde{N} = 256$
ECB-IR2	20	10	5	5	10	10	196	$\tilde{N} = 256$

differences and relatively insensitive to small differences between two modes. Another advantage is that MAC produces a real quantity despite the degree of complexity of the modes. Considering MAC as a correlation constant we can find:

$$\text{MAC}_{rs} = \frac{|\phi_r^H \phi_s|^2}{(\phi_r^H \phi_r)(\phi_s^H \phi_s)}, \quad (34)$$

where ϕ_r and ϕ_s are two mode vectors to be compared and a superscript H denotes the Hermitian transpose, [42]. The MAC yields a normalised correlation coefficient between two modes with a value between zero and unity, where unity denotes excellent correlation and zero denotes absolute orthogonality.

4.1. Stiffened steel plate

In the first example, a stiffened plate inspired by [43] made of steel is considered. Such plates are on a larger scale commonly employed in bridge structures to withstand lateral torsional buckling as a consequence of significant compression perpendicular to its axial direction. The dimension of the plate is as follows, Length L is 4.8 [m], width B is 3.2 [m] and thickness t is 0.03 [m]. The plate is stiffened in the axial and transversal directions with two and four stiffeners, respectively. The stiffeners in the axial direction have a centre-to-centre distance of 1.2 [m], whilst the transversal stiffeners have a centre-to-centre distance of 1 [m]. The stiffeners have a height H of 0.5 [m]. To the plate, material properties of steel are assigned, with Young's modulus E as 210 [GPa], density ρ as 7850 [kg/m³] and Poisson's ratio of 0.3. The domain is discretised using shell FEs with 6 DOFs per nodal point and linear approximation as well as reduced integration. The element size of 0.1 [m] is chosen in each direction resulting in 2656 elements with 2727 nodes and a total number of DOFs of 16362. The system matrices are subsequently derived and partitioned to 6 substructures coupled at the interface Γ denoted by a blue line in Fig. 2. Information regarding the number of modes employed in the truncation of internal and interface dynamics of each region can be found in Table 2. Throughout the analyses, no Dirichlet boundary condition has been imposed on the plate and no rigid-body modes and resonances have been included in the results presented herein.

The evaluation results indicate that the interface reduction technique IR2 in conjunction with ECB, i.e. ECB-IR2, performs better than the classical interface reduction technique using CB-IR1 and ECB-IR1, in terms of accuracy in the prediction of the resonance frequencies, Fig. 3. The convergence plot in Fig. 4 shows that CB-IR1 converges faster than ECB-IR1 and ECB-IR2. This suggests that adding more interface modes in CB-IR1 does not enhance the results, while the opposite is true for ECB-IR1 and ECB-IR2. Thus, users might prefer to select the convergence criteria in ECB-IR2, cf. Table 2 and Fig. 4. Evaluation of the transfer functions indicates improvement in accuracy when ECB-IR2 is employed compared to CB-IR1 and ECB-IR1, Fig. 5. Notably, the ECB-IR2 with only 256 DOFs produces normal modes with an accuracy similar to the ECB with 1032 DOFs in comparison with the full model with 16362 DOFs, Fig. 6.

4.2. Composite cross-laminated timber plate

Cross-laminated timber (CLT) plates have gained great popularity as a sustainable alternative to conventional construction materials, such as steel and concrete, since the commercial launch of engineered wood products (EWPs). However, in contrast to the mentioned conventional construction materials, wooden structural elements possess a low mass density, making them extra sensitive to vibration at lower frequencies. The structure of CLT consists of an odd number of wooden boards arranged in layers, with each layer oriented at a 90-degree angle to the adjacent ones. For additional information regarding CLT and its production and development, the reader is referred to the state-of-the-art review by Brandner [44] and the overview in [45]. Wood as a material possesses a high degree of orthotropy in three main directions: longitudinal (parallel to wood fibres), denoted as L ; tangential (tangent with the concentric growth rings), denoted as T ; and radial (perpendicular to the concentric growth rings), denoted as R .

In this example, a CLT plate with a total length of 7 [m] and a width of 2.4 [m] is considered, Fig. 7. The plate consists of 7 cross-wisely bonded layers with a material orientation of 90° about the z -axis between one layer and the adjacent ones. The plate is discretised using doubly-curved composite shell elements with 7 layups, [47]. The thickness of each layup is 40| 20| 40| 40| 40| 20| 40 millimetres, respectively. For the bottom and top layers, the material direction of the fibres is spanning along the x -axis, whilst for the second layer from the bottom the fibres are oriented by an angle of 90° about the z -axis and so forth. Such plates are commonly used as structural elements in multi-storey timber buildings. To the plate, orthotropic material properties of softwood of strength

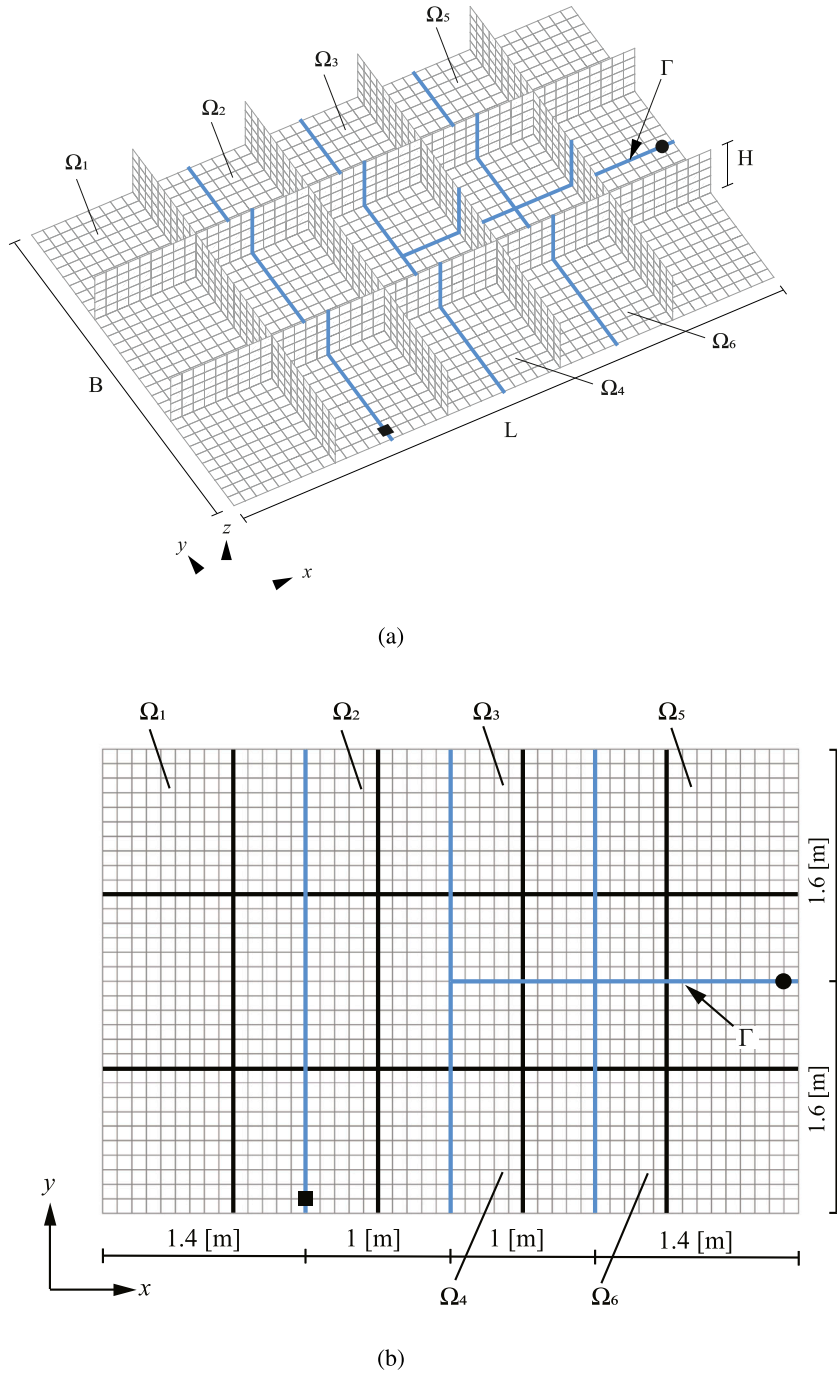


Fig. 2. Discretised geometry of a stiffened plate with $L = 4.8$ [m], $B = 3.2$ [m], $H = 0.5$ [m] in (a) 3D and (b) in 2D, as discussed in Section 4.1. Black lines in (b) denote the stiffeners and $\Omega_i \in \{1, 2, \dots, 6\}$ denote substructure regions. The blue line Γ denotes the interface, a black circle denotes the unit excitation point and a black square denotes the evaluation point of the transfer functions in the frequency domain.

class C35 in [46] have been assigned, Table 3. Throughout the analyses, no Dirichlet boundary condition has been imposed on the plate and the rigid-body resonances and modes have not been considered in the results presented herein. Further information regarding the number of modes retained for each substructure can be found in Table 4.

The evaluations indicate improvement in the accuracy prediction of the resonance frequencies associated with the ECB-IR2 compared with CB-IR1 in comparison to the full model, Fig. 8. The interface mode convergence indicates that CB-IR1 and ECB-IR2

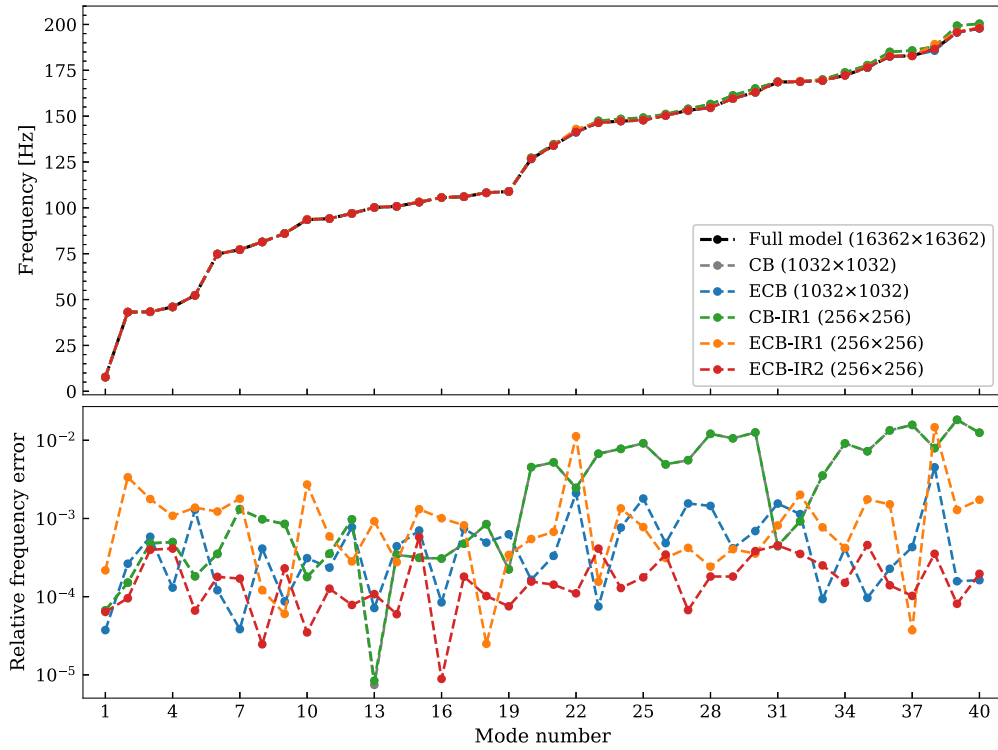


Fig. 3. Resonance frequencies and their associated relative error of the stiffened plate example in Section 4.1. The numbers in the legend indicate the size of the system matrices.

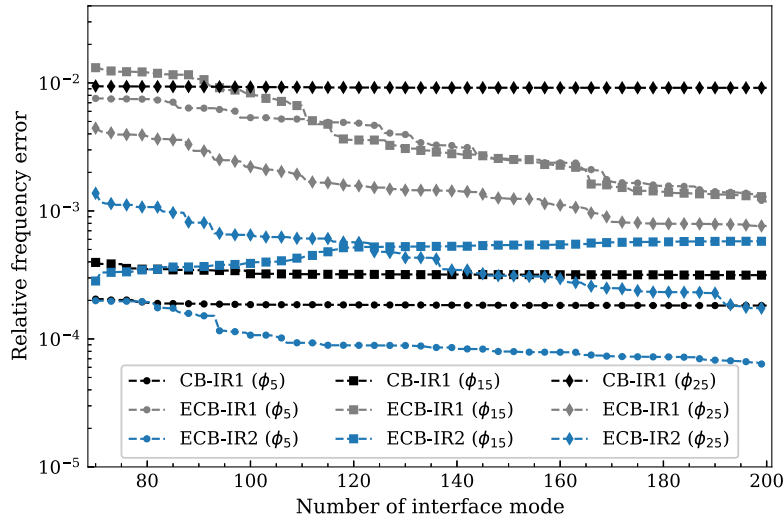


Fig. 4. Interface mode convergence of the stiffened plate example in Section 4.1, with respect to modes number 5, 15, and 25.

Table 3

Orthotropic material properties of CLT of strength class C35 from Ref. [46]. The subscripts L , T , and R respectively denote orthotropic orientations of wood in longitudinal, tangential and radial directions.

E_L [MPa]	E_T [MPa]	E_R [MPa]	G_{LT} [MPa]	G_{LR} [MPa]	G_{RT} [MPa]	ν_{LT} [-]	ν_{LR} [-]	ν_{RT} [-]	ρ [kg/m ³]
13 000	262	430	810	810	57	0.48	0.42	0.28	470

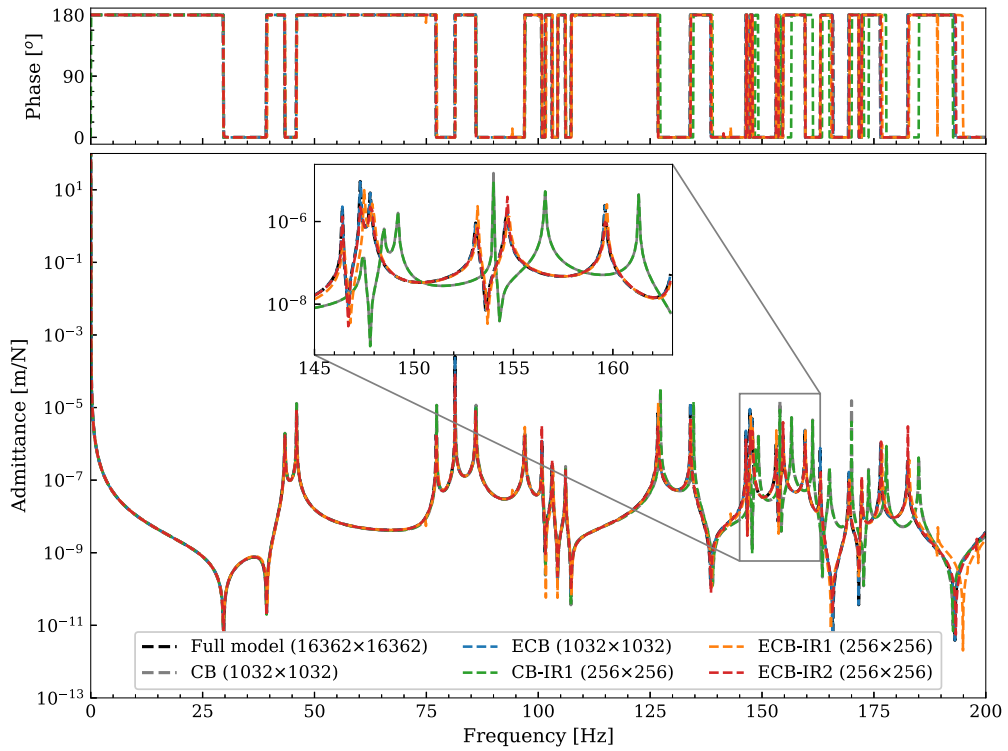


Fig. 5. Bode plot of the transfer functions and their associated phase shift angle of the stiffened plate example in Section 4.1, to unit loading. The numbers in the legend indicate the size of the system matrices.

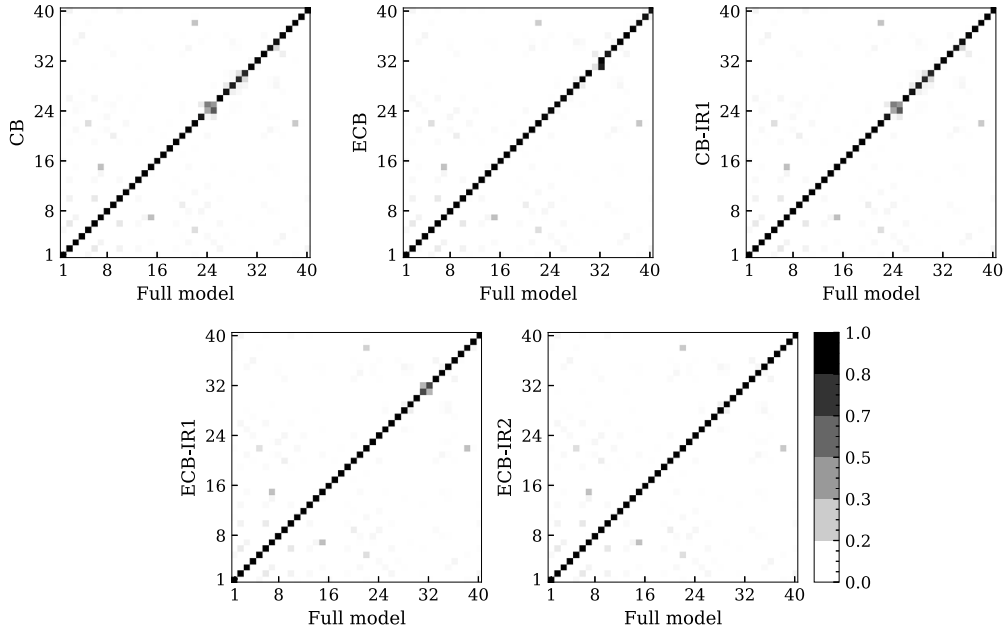


Fig. 6. Modal Assurance Criterion comparing 40 flexible modes of the stiffened plate example in Section 4.1.

converge faster than the ECB-IR1, Fig. 9. This explains that an increase in the number of interface modes in CB-IR1 does not improve its accuracy, while the opposite is true for ECB-IR1 and ECB-IR2. Evaluation of the transfer functions indicates further improvement in predicting the resonance peaks in Fig. 10. The MAC evaluation also indicates that ECB-IR2 produces normal modes with only 110 DOFs with an accuracy nearly as for the ECB with 281 DOFs in comparison to the full model with 2808 DOFs, Fig. 11. To gain

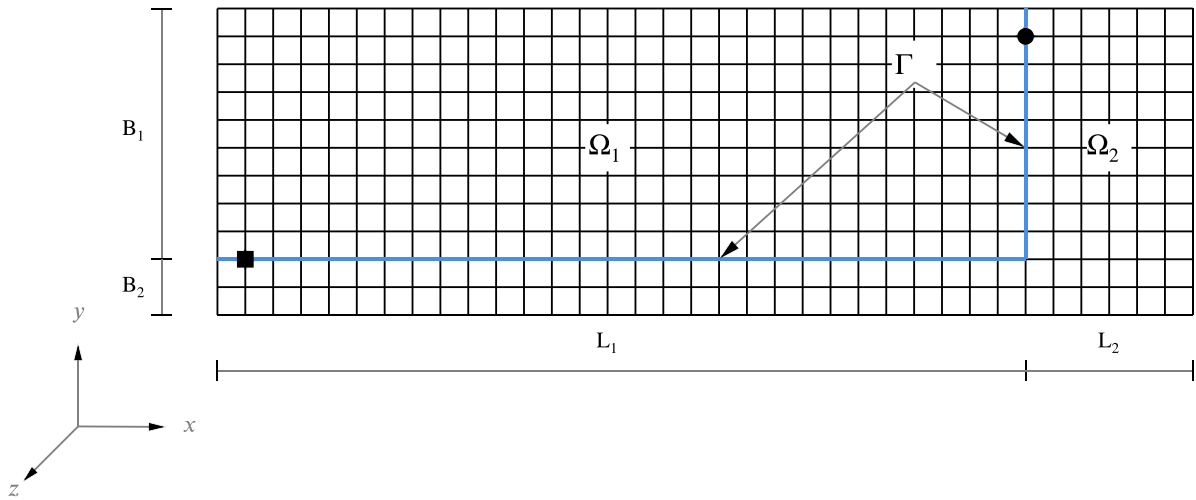


Fig. 7. Discretised Composite Cross-Laminated Timber (CLT) plate geometry (Section 4.2): Substructures Ω_1 and Ω_2 are interface-coupled at Γ . A black circle represents unit loading in the out-of-plane direction (z-axis), while a black square indicates the corresponding transfer function. Geometrical dimensions: $L_1 = 5.8$ [m], $L_2 = 1.2$ [m], $B_1 = 2$ [m], and $B_2 = 0.4$ [m].

Table 4

Number of internal and interface modes computed for the composite CLT plate example in Section 4.2.

Method	Number of internal modes retained		Number of interface modes retained	Total number of DOFs
	$N_{i,d}^{(1)}$	$N_{i,d}^{(2)}$		
Full model	1740	828	240	$N = 2808$
CB	25	16	240	$\tilde{N} = 281$
ECB	25	16	240	$\tilde{N} = 281$
CB-IR1	25	16	69	$\tilde{N} = 110$
ECB-IR1	25	16	69	$\tilde{N} = 110$
ECB-IR2	25	16	69	$\tilde{N} = 110$

further insight into the modes predicted by each model in this study, 3D visualisations of modes 13 and 26 are shown in Figs. 12 and 13.

4.3. Multi-storey CLT building

In the last example a multi-storey CLT building comprising 4 CLT plates, a CLT core and 18 Glued-Laminated Timber (GLT) beams, is fragmented into 4 substructures. Each substructure is connected at the interface Γ , indicated by a blue line in Figs. 14 and 15. The plane dimension (length $\times$ breadth) of the building is 20×16 [m²] and each storey is connected with another at a height of 3.4 [m], giving the building a total height of 13.6 [m]. The core has the dimension (length $\times$ breadth) of 12×8 [m²], placed at the centre of the building. The beams have a cross sectional area of 215×405 [mm²] with a centre-to-centre distance of 4 [m]. The core and the plates are assumed to be of 5-ply CLT elements with layup thicknesses 40| 40| 40| 40| 40 millimetres and strength class of C35, for which the orthotropic elastic properties are presented in Table 3. For the plates, the top and bottom layers of the CLT plates span along the global x – axis, whilst for the core, the top and bottom layers of the CLT span along the global z – axis. Elastic properties of GL30c with Young's modulus of 13000 [MPa], density of 500 [kg/m³] and Poisson's ratio 0.42 have been assigned to the beams.

The CLT core and plates are discretised using doubly curved composite shell elements with reduced integration and hourglass control, as described in [47]. The global element size is set to 0.5 [m] in each direction, using linear interpolation. Additionally, a convergence analysis of the discretisation has been performed. The beams are modelled according to the Bernoulli–Euler beam theory (without shear deformation) with the same element size as the plates and the core. Full interaction between the upper and lower floors as well as the beams and the floors is considered. The geometry of the building analysed herein is inspired by the study in [48]. Throughout the analyses presented herein, pinned (fixed translational DOFs) Dirichlet boundary conditions were applied to the bottom of the multi-storey building, i.e., the bottom of the beams and the core. The rigid-body resonances and their associated modes have not been considered in the results presented herein. The number of modes computed for each model can be found in Table 5.

The evaluation results align with the previously presented examples, demonstrating that ECB-IR2 provides resonance frequencies with enhanced accuracy compared to CB-IR1 and ECB-IR1. This improvement is particularly evident in the prediction of the lower

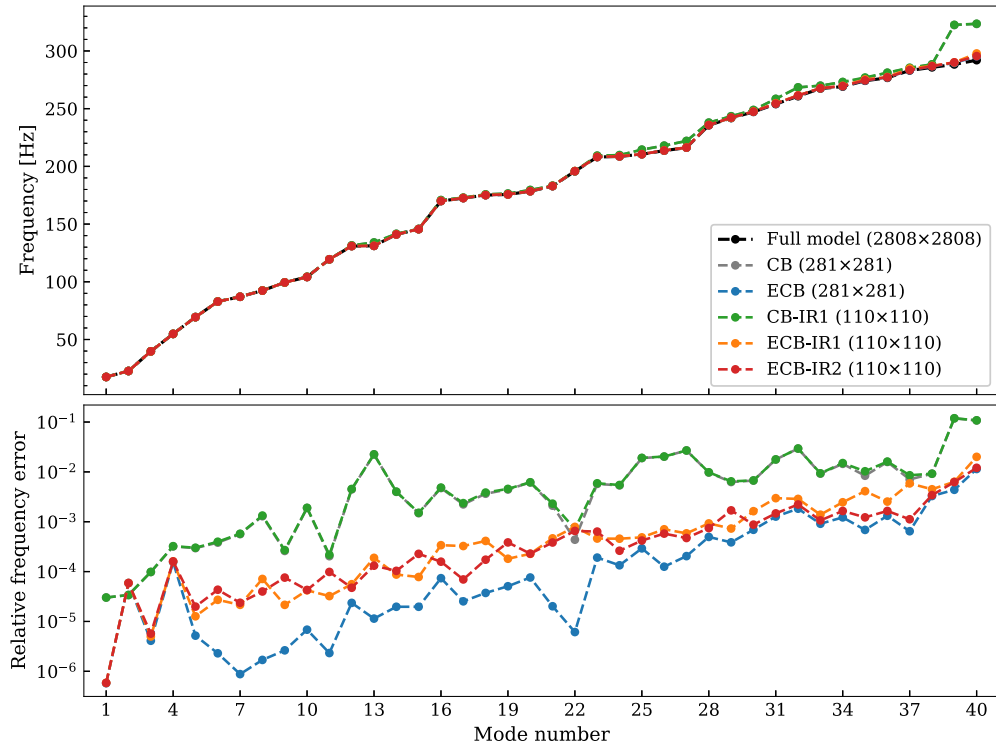


Fig. 8. Resonance frequencies and their associated relative error of the composite CLT plate example in Section 4.2. The numbers in the legend indicate the size of the system matrices.

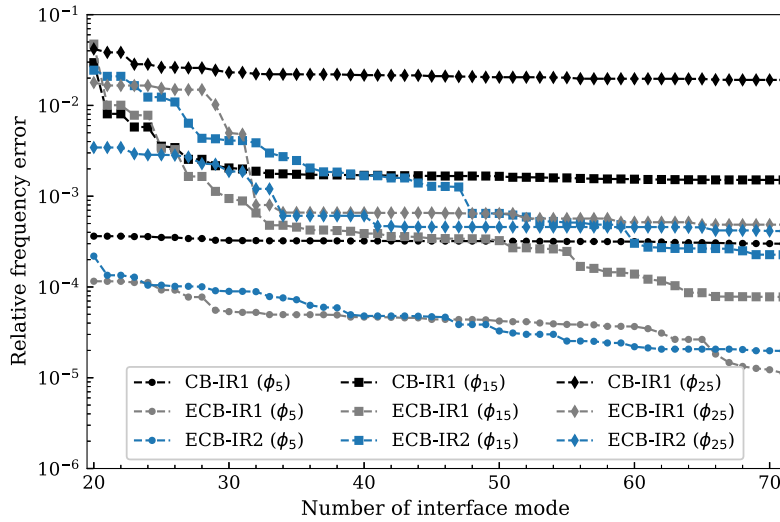


Fig. 9. Interface mode convergence of the composite CLT plate example in Section 4.2, with respect to modes number 5, 15, and 25.

frequency elastic resonances, as shown in Fig. 16. The interface mode convergence analysis in Fig. 17 indicates that increasing the number of modes further can enhance the accuracy of ECB-IR1 and ECB-IR2. Therefore, users may choose a convergence criteria that meets their requirements. In the prediction of the transfer functions, an obvious improvement can be observed in the frequency range of 9–11 [Hz], in particular the double peaks, Fig. 18. Thus the transfer functions predicted by ECB-IR2 are of greater accuracy compared to CB-IR1 and ECB-IR1, in comparison to the full model. Furthermore, the MAC evaluations indicate that the ECB-IR2 with

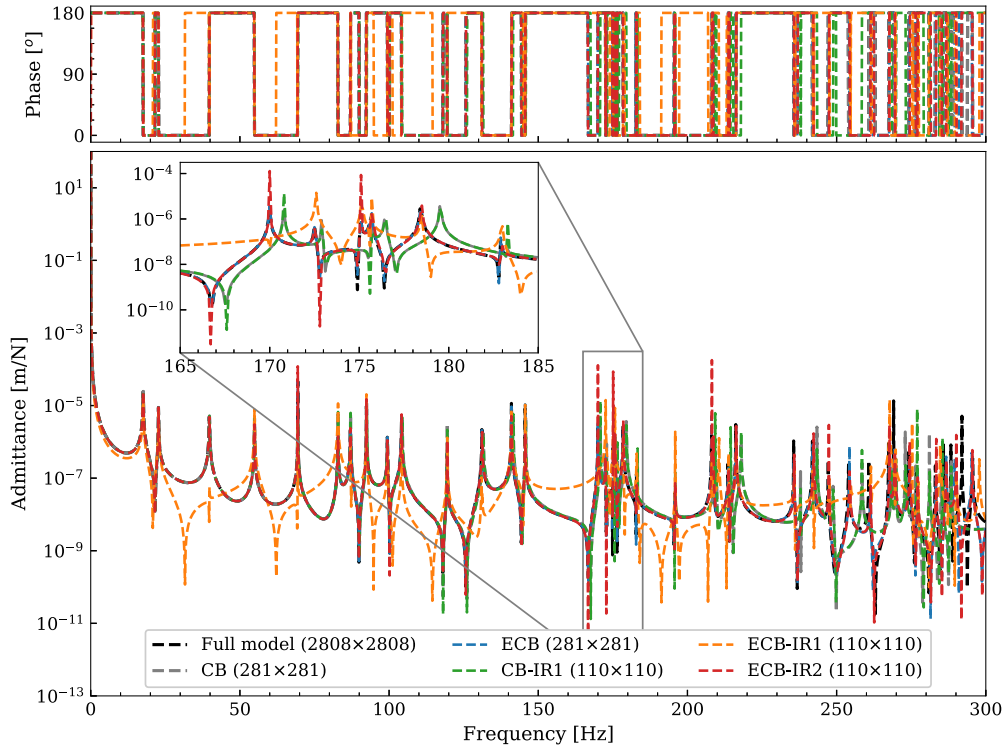


Fig. 10. Bode plot of the transfer functions and their associated phase shift angle of the composite CLT plate example in Section 4.2, to unit loading. The numbers in the legend indicate the size of the system matrices.

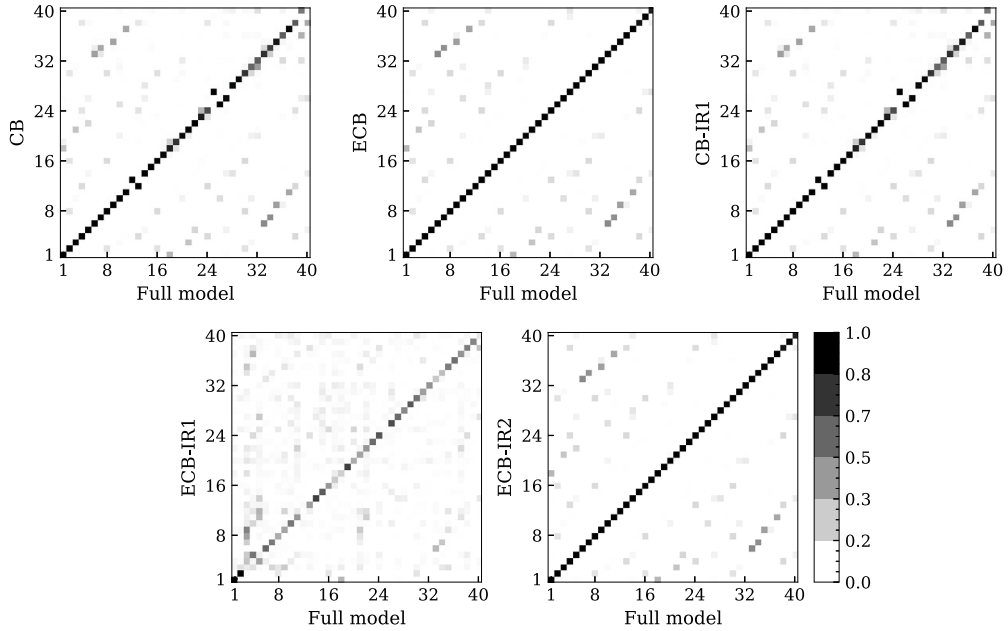


Fig. 11. Modal Assurance Criterion comparing 40 flexible modes of the composite CLT plate example in Section 4.2.

only 342 DOFs produces vibration modes nearly as accurate as the ECB method with 1559 DOFs in comparison with the full model with 38892 DOFs, Fig. 19. Consistent with previous results, ECB-IR1 does not achieve the same level of accuracy in normal modes as CB-IR1 or ECB-IR2. This discrepancy can be attributed to the influence of internal and interface stiffness coupling characteristics of the ECB method, Fig. 19.

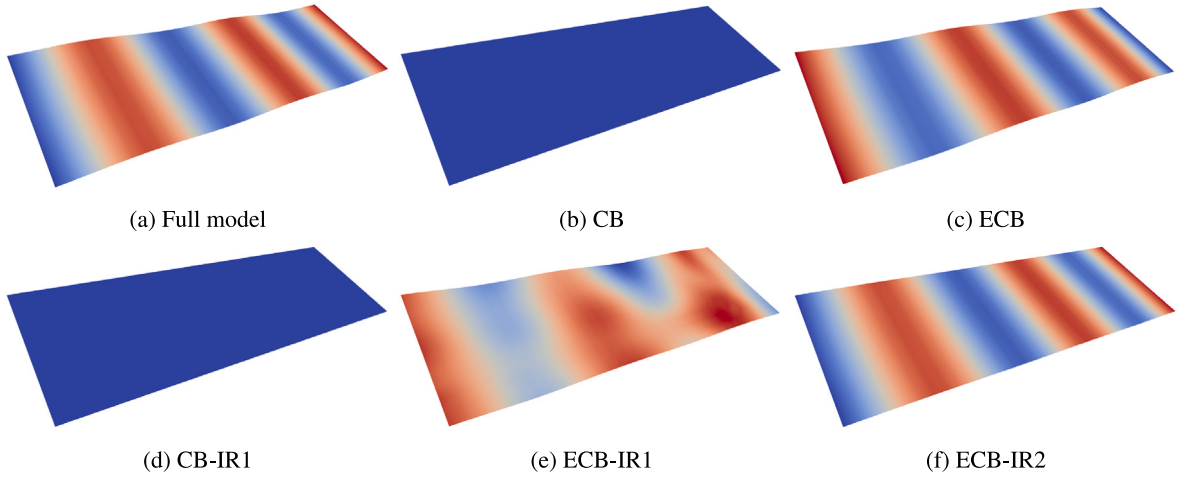


Fig. 12. 3D visualisation of 13th flexible mode of the CLT plate discussed in Section 4.2.

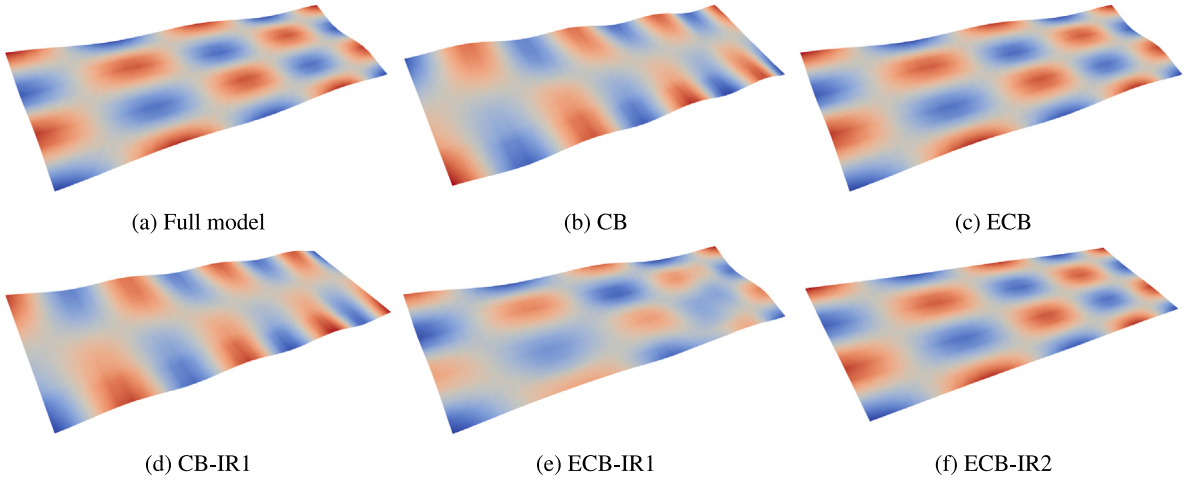


Fig. 13. 3D visualisation of 26th flexible mode of the CLT plate discussed in Section 4.2.

Table 5

Number of internal and interface modes computed for the multi-storey building example in Section 4.3.

Method	Number of internal modes retained				Number of interface modes retained	Total number of DOFs
	$N_{i,d}^{(1)}$	$N_{i,d}^{(2)}$	$N_{i,d}^{(3)}$	$N_{i,d}^{(4)}$		
Full model	7818	13 974	10 896	4740	1464	$N = 38892$
CB	15	20	35	25	1464	$\hat{N} = 1559$
ECB	15	20	35	25	1464	$\tilde{N} = 1559$
CB-IR1	15	20	35	25	247	$\check{N} = 342$
ECB-IR1	15	20	35	25	247	$\tilde{N} = 342$
ECB-IR2	15	20	35	25	247	$\tilde{N} = 342$

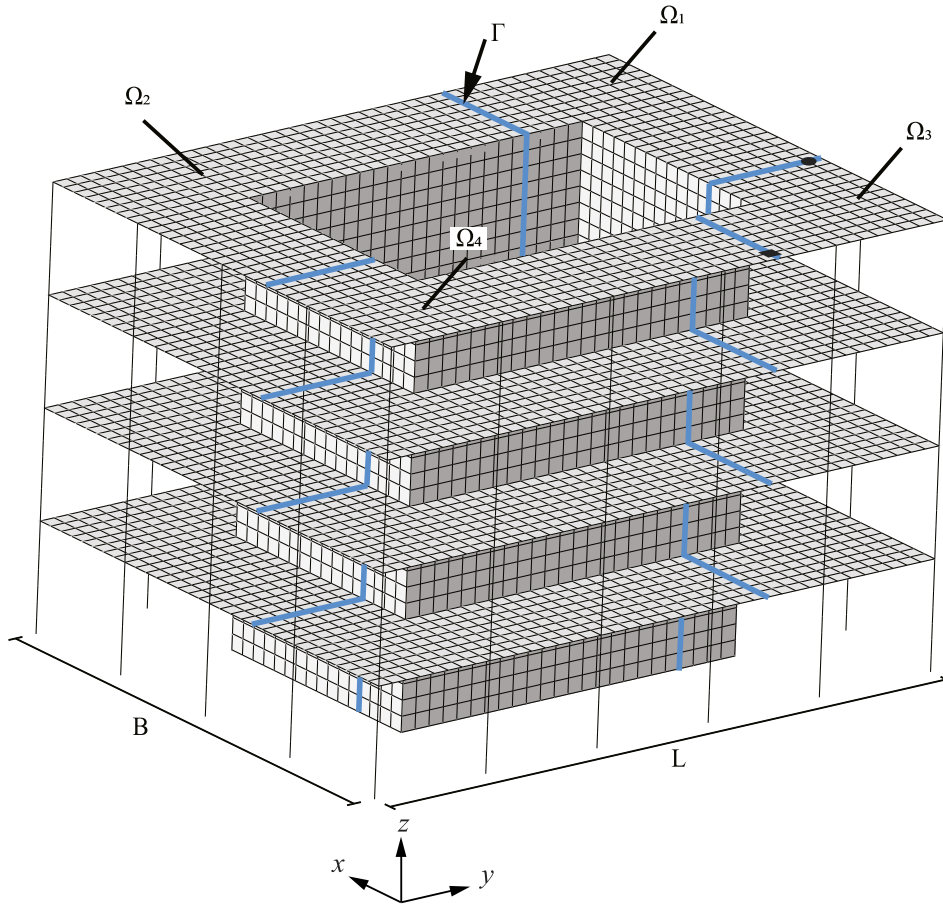


Fig. 14. Discretised geometry of a multi-storey timber building with GLT supporting beams and 5-ply CLT core and plates in Section 4.3. Geometrical dimensions length (L) $\times$ breadth (B) $\times$ height (H) are $20 \times 16 \times 13.6$ [m³]. The notation $\Omega_i \in \{1, 2, \dots, 4\}$ denotes substructure regions, whilst a blue line denotes the coupling interface Γ . A black circle denotes unit loading, and a black square denotes the corresponding transfer function.

5. Conclusions

In this study, two interface reduction techniques based on Characteristic Constraint (CC) modes in conjunction with Craig-Bampton (CB) and Enhanced Craig-Bampton (ECB) methods were presented. Among these techniques, ECB-IR2, which considers coupling between truncated internal and interface DOFs, outperforms the classical method proposed by Castanier et al. [30] when ECB is employed. The coupling term is particularly important due to the stiffness coupling characteristics of the ECB method. This approach provides an improved representation of vibration energy transmission between the truncated internal and interface dynamics. This improves accuracy in predicting resonance frequencies, transfer functions, and normal modes. The ECB-IR2 is a valuable choice when the accuracy of reduced system matrices is of great importance.

The application examples suggest that models with reduced interface dynamics achieve accuracy levels comparable to their CB and ECB counterparts (without interface reduction), from which they are derived. Notably, the inclusion of a large number of CC modes did not enhance the accuracy of the CB-IR1 method, contrasting with the ECB-IR1 and ECB-IR2, where additional CC modes improved accuracy. Therefore, it is evident that additional CC modes enhance the accuracy of the ECB method with interface reduction, especially in the case of ECB-IR2. Furthermore, CB-IR1 demonstrated better convergence properties, while ECB-IR2 exhibited better accuracy with the same number of CC modes.

Potential future avenues of study may include extending the present research to interface reduction of free-interface CMS models, such as Rubin's approach in [14] and the Dual Craig-Bampton method by Rixen [39]. Additionally, exploring the performance of ECB-IR2 in coupled vibroacoustic problems, multi-physics scenarios, and systems involving heat, mass transfer, and damping could be of interest.

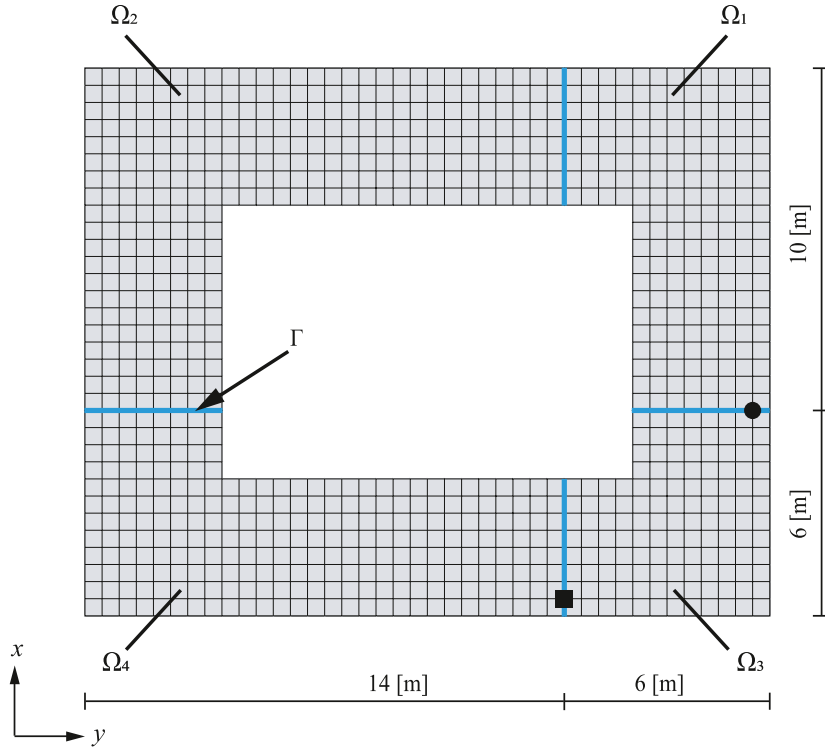


Fig. 15. Discretised plate geometry of a 5-ply CLT plate with length (L) $\times$ breadth (B) of 20×16 [m²], in Section 4.3. The inner core has dimensions length $\times$ breadth of 12×4 [m²]. An Ω denotes a substructural region, whilst a blue line denotes coupling interface Γ . The black circle and rectangle denote respectively excitation node and evaluation node in the frequency domain.

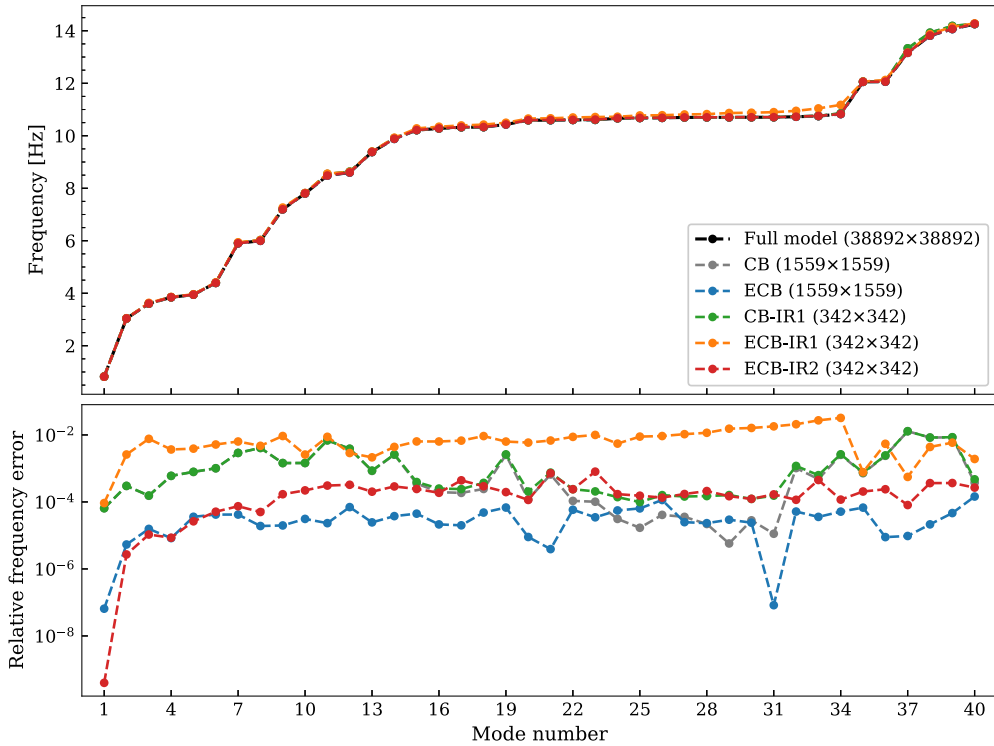


Fig. 16. Resonance frequencies and their associated relative error of the multi-storey CLT building example in Section 4.3. The numbers in the legend indicate the size of the system matrices.

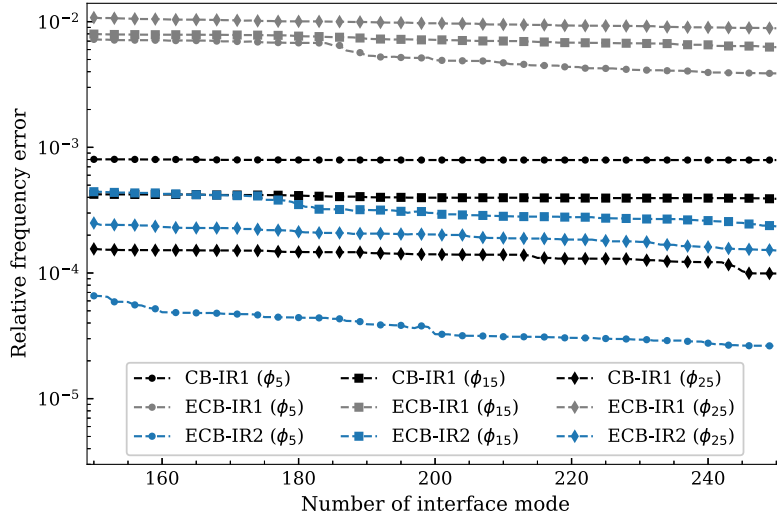


Fig. 17. Interface mode convergence of the multi-storey CLT building example in Section 4.3, concerning modes number 5, 15, and 25.

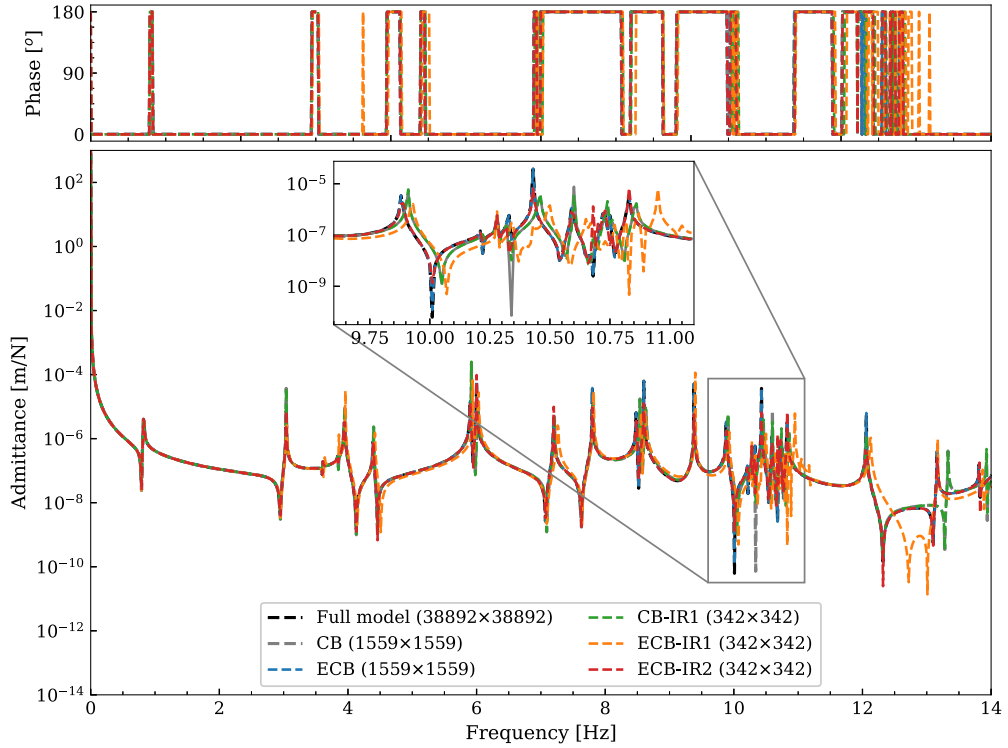


Fig. 18. Bode plot of the transfer functions and their associated phase shift angle of the multi-storey CLT building example in Section 4.3, to unit loading. The numbers in the legend indicate the size of the system matrices.

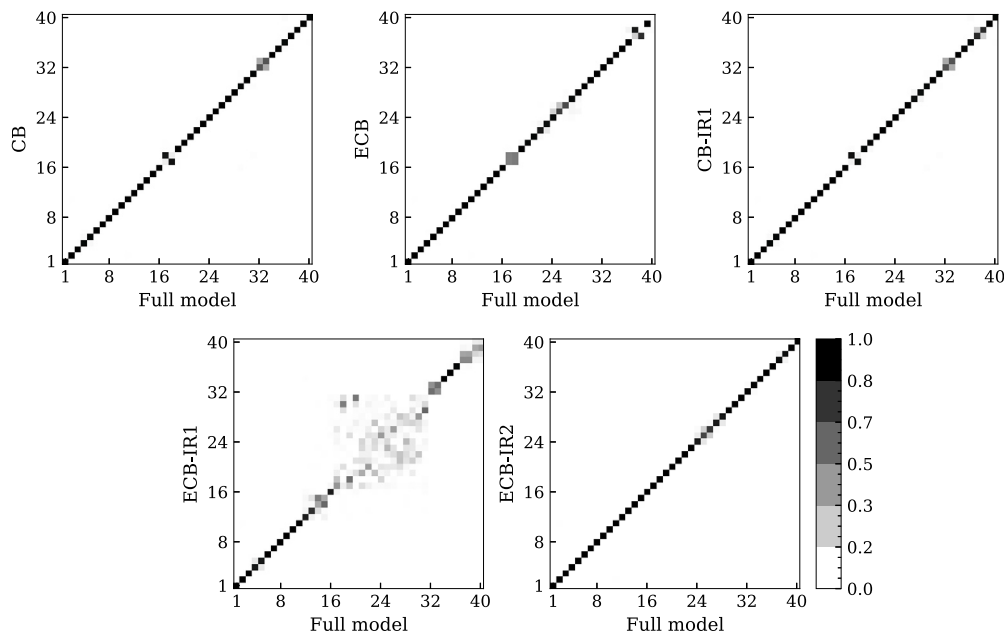


Fig. 19. Modal Assurance Criterion comparing 40 flexible modes of the multi-storey CLT building example in Section 4.3.

CRediT authorship contribution statement

Benjamin Bondsman: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Chang-uk Ahn:** Conceptualization, Formal analysis, Methodology, Validation. **Jin-Gyun Kim:** Conceptualization, Funding acquisition, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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